Learning Verified Monitors for Hidden Markov Models

Abstract. Runtime monitors for hidden Markov models obtain a sequence of observations and must classify whether the system is currently deemed safe. A system is deemed safe, if the probability of being in a dangerous state conditioned on these observations is below some threshold. The key contribution is the first correct-by-construction synthesis method for such monitors, represented as finite state machines. The contribution combines four ingredients: First, we establish the coNP-hardness of checking whether a finite state machine is a correct monitor, i.e., a monitor without misclassifications. Second, we provide a reduction that reformulates the search for misclassifications into a standard probabilistic system synthesis problem. Third, we integrate the verification routine into an active automata learning routine to synthesise correct monitors. Fourth, we provide a prototypical implementation that shows the feasibility and limitations of the approach on a series of benchmarks.

1 Introduction

Runtime assurance is an essential ingredient in the deployment of safe autonomous systems [**xx**]. Runtime monitors provide assurance by flagging potentially dangerous system behavior by observing a system execution. More precisely, a monitor receives a stream of observations about the system and outputs a verdict, e.g., it raises an alarm that the system has left some safety envelope. A monitor is correct if it correctly raises such alarms based on a formal specification. Various challenges in creating correct runtime monitors for (semi-)autonomous systems have been identified [32], such as: (1) the state of the system is only partially observable, i.e., the stream of observations comes from sensor readings and does not uniquely identify the state of a system, (2) the behavior of the system and/or the sensors may be subject to probabilistic uncertainty, (3) the monitor itself is subject to resource constraints (time, memory, etc), and (4) the monitor is itself safety-critical and should therefore be subject to extensive validation. Challenges (1,2) can be addressed by modelling the system as a hidden Markov model (HMM), Challenges (3,4) can be addressed by representing a monitor as, e.g., a (small) finite state machine. Concretely, this paper focuses on the following question: *Is a given finite state machine a correct monitor for a given and known HMM?* This paper studies the complexity of this problem, provides a practical verification approach, and embeds that approach into a framework to learn monitors.

What are HMMs? In this paper, we assume that the system including its sensors is adequately described as a discrete Hidden Markov Model (HMM) [**xx**] and that we have full access to this HMM. Markov chains (MCs) describe system

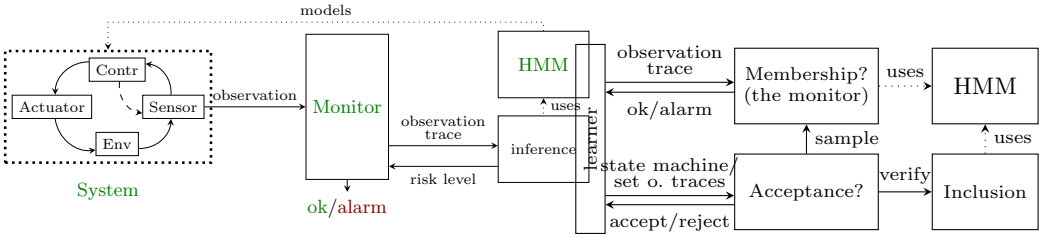


Fig. 1: White-box monitoring of systems

Fig. 2: Learning monitors.

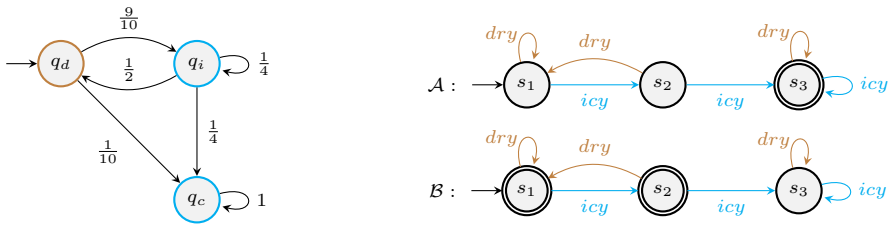
behavior subject to probabilistic uncertainty. Paths through an MC are sequences of states that describe system executions. HMMs extend MCs by labelling their states with *observations*. Intuitively, the observations can be used to model the information that the monitor receives in every state. In HMMs, every path can be lifted to a sequence of observations, which we call a *trace*. The trace associated to a system execution describes the information received by the monitor.

Monitoring with HMMs. Monitoring with HMMs assumes that a monitor receives a trace from the system and uses inference on the HMM modelling the system (see Fig. 1). In the inference step, the key task is estimate whether the current system state is dangerous, based on the available information in form of a trace. Intuitively, the *risk of a trace* [27] quantifies how likely it is that the system state is dangerous. Formally, this can be defined as the probability of ending in a dangerous state, conditioned on the fact that the system execution matches the trace. For a given trace, we may compute this risk, e.g., either via model checking [13] or by a (forward) filtering that tracks a distribution over the current states [**xx**]. We call a trace *unsafe* if its risk exceeds an acceptable threshold. Monitors should raise alarms only for unsafe traces.

What are correct monitors? Like in [1], we summarise the behavior of monitors by the set of traces (i.e., formal languages) on which they raise an alarm. A monitor is correct iff it raises an alarm on all unsafe traces. We highlight that a monitor can be correct without doing inference at run time [17]! The key verification problem in this paper asks whether a monitor, represented as a deterministic finite automaton, accepts (i.e., raises an alarm) exactly the unsafe traces. In this paper, we only consider traces that are bounded by some fixed horizon.

Illustrative toy example. As an running example, we consider an oversimplified car driving scenario, loosely inspired by runtime monitors obtained from high-fidelity simulations [37]. A car can be in three states: It can be on a dry road, on an icy patch, or it can drifted off the road. The HMM in Figure 3 describes how a car alternates between dry and icy road segments, and where being on a icy (dry) segment positively affects the probability that the next segment is icy (dry). When on an icy road, there is a higher probability to drift off the road. Our sensor can detect dry roads, but cannot distinguish between icy roads and off-road conditions. We can define the trace risk as the probability to be off-road, conditioned on a trace, such as $dry \cdot icy \cdot icy$. In Fig. 3b, monitor $\mathcal{A}$ is a finite

SJ: Can we give the trace risk here, as in, can we say how likely it is that this trace means being offroad?



(a) HMM. States q_i, q_c have the observation *icy*, state q_d has *dry*.

(b) A correct monitor $\mathcal{A}$ and an incorrect monitor $\mathcal{B}$ (for $\lambda = 0.8$).

Fig. 3: Running example: HMM (a) and two monitors (b).

automaton that describes the set of traces that end with two consecutive icy patches. The key verification question in this paper is now: Is this a correct monitor for threshold 0.9? Put differently, does it raise an alarm iff the trace risk is above 0.9?

SJ: caption requires a hrizon?

Computational complexity results. Deciding whether a monitor is correct is coNP-complete (Theorems 4), assuming that the horizon is unary encoded. In particular, we study the dual problem that asks whether a monitor misclassifies at least one trace as safe/unsafe. A simpler variant of this problem asks whether there is a bounded trace in the HMM that is unsafe. This problem is already strongly NP-hard (Lemma 5) and the optimization problem that asks to compute (the risk of) most risky trace is APX-hard (Lemma 6), which indicates that it is intractable to even approximate this risk.

Our verification approach. While deciding whether a monitor is correct is in general not tractable, we suggest to utilise recent advances in the synthesis of probabilistic systems [6]. We concentrate on proving the absence of *missed alarms*, i.e., we concentrate on showing that the monitor correctly identifies every unsafe trace. A similar reduction works to show that a monitor correctly identifies safe traces. First, for a single trace, computing the risk can be reduced to computing a reachability probability in a MC that is some kind of product between a DFA that accepts exactly the trace and the HMM [27]. Thus, inspired by [11], we reduce our problem to the question: *Is there a DFA (accepting exactly one trace, accepted by our monitor) such that the probability in the product MC exceeds a threshold?* The answer is no iff the monitor has no missed alarms. We formalise the adequate problem using colored MDPs and solve it using (exact) probabilistic model synthesis methods [5] as implemented in the tool Paynt [6].

Our learning approach. Ultimately, we do not only want to verify the correctness of a monitor, but we want to *synthesise* such monitors. Above, we already mentioned that the monitors are formal languages. When considering bounded traces, these monitors are regular, and can thus be captured using DFAs. Thus, we aim to synthesise DFAs. We choose to do this using active automata learning (AAL, [7, 38]), which is similar to other popular oracle-guided inductive synthesis loops [**DBLP:journals/acta/JhaS17**]. To use AAL, we must pro-

vide a *membership oracle* that decides for a single trace whether the risk of the trace exceeds a threshold and an *equivalence oracle* that decides whether a given hypothesis monitor is indeed correct. The membership oracle can be implemented via inference for standard monitors on HMMs (see above), while the equivalence oracle can be implemented by verifying the correctness of the monitor. Practically, some minor modifications are necessary to embed our approach in AAL framework: We have to handle the finite horizon, resolve some traces that are not relevant for the monitor, and we combine conformance and equivalence queries to boost performance.

Contributions. In summary, this paper provides a first framework to learn runtime monitors — encoded as finite automata — that are verified to be correct with respect to a given HMM. The main contributions are: (1) We solve the verification problem by exhaustively searching for a counterexample using probabilistic system synthesis (Sec. 3). (2) We learn monitors by using the verifier above to answer equivalence queries in conjunction with active automata learning (Sec. 4). (3) We prove the hardness of the verification problem (Sec. 5). (4) We demonstrate the feasibility and limitations of both the verifier and the learner on a series of benchmarks (Sec. 6) Our prototype can handle HMMs with thousands of states, up to a hundred observations, and monitors with over 100 states.

2 Formal Problem Statement

Let X be a finite set. A distribution μ over X is a mapping $\mu\colon X \to [0,1]$ such that $\sum_{x \in X} \mu(x) = 1$. The set of all distributions over X is written $\Delta(X)$.

DFAs. A *deterministic finite automaton* (DFA) is a tuple $\mathcal{A} = (Q, \Sigma, \delta, \iota, F)$. Q is a finite set of *states*, Σ is an *alphabet*, $\delta\colon Q \times \Sigma \rightharpoonup Q$ is a (partial) *transition function*, ι is the *initial state*, and $F \subseteq Q$ is a set of *accepting* states[1]. Let ε denote the empty word. We lift the transition relation to words, which is defined recursively: $\delta^*(q, \varepsilon) = \varepsilon$ if $a \in \Sigma$ and $\delta^*(q, w \cdot a) = \delta(\delta^*(q, w), a)$. The *language* $\mathcal{L}(\mathcal{A})$ of a DFA $\mathcal{A}$ is the set of all words that end in a final state of $\mathcal{A}$, $\mathcal{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \delta^*(\iota, w) \in F\}$. We say that $\mathcal{A}$ accepts $w \in \mathcal{L}(\mathcal{A})$.

Luko: Do we actually use this?

2.1 Models

We introduce MDPs and HMMs: The former are integral to our approach, and the latter are crucial to the problem statement. Details can be found in [12].

Definition 1 (MDP). *A* Markov decision process *(MDP) is a tuple* $(S, \iota, Act, \mathbf{P})$ *with a countable nonempty set* S *of* states*, the* initial state $\iota \in S$*, the* partial transition function $\mathbf{P}\colon S \times Act \rightharpoonup \Delta(S)$.

We use $Act(s) = \{a \mid \mathbf{P}(s, a) \neq \bot\}$ as the set of *enabled actions*. We assume no deadlocks, i.e., for every $s \in S$, $Act(s) \neq \emptyset$. A path π is a sequence $s_0 a_0 s_1 \ldots \in$

[1] F is named after the synonymous final states to avoid confusion.

$(S \times Act)^* \times S$, such that $a_i \in Act(s_i)$ and $P(s_i, a_i)(s_{i+1}) > 0$ for every $i \geq 0$. The last state of a finite path is denoted by $\pi_\downarrow$. The set of paths in MDP $\mathcal{M}$ is denoted as $\Pi^{\mathcal{M}}$, the set of finite paths is $\Pi^{\mathcal{M}}_{fin}$, the set of paths of at most length h are $\Pi^{\mathcal{M}}_{h}$, and the set of paths of exactly length h are $\Pi^{\mathcal{M}}_{=h}$.

A *Markov chain* (MC) is an MDP where $|Act(s)| = 1$ for every state $s \in S$. We simplify notation and write MCs as a tuple $(S, \iota, \mathbf{P})$ and write $\mathbf{P}(s)$ to refer to the unique distribution $\mathbf{P}(s, a)$. Paths in a MC can be simplified to be sequences of (only) states. The probability measure $Pr^{\mathcal{M}}$ of an MC $\mathcal{M}$ is the unique probability measure following from the canonical σ-algebra associate with $\mathcal{M}$. A reachability property $\Diamond T$ on a set of target states T, is the set of paths which contain a state $t \in T$, and is defined using the standard cylinder set construction.

Definition 2 (HMMs). *A (risk-labelled) HMM is a tuple $(S, \iota, \mathbf{P}, Z, obs, r)$ such that $(S, \iota, \mathbf{P})$ is a MC, Z is a finite set of observations, and $obs\colon S \to Z$ is the (deterministic) observation function*[2]*, and $r\colon S \to \mathbb{R}_{\geq 0}$ is the risk function.*

Notions such as paths are lifted from MCs. Furthermore, traces τ are sequences of observations, and the set of traces of a HMM, for which there is non-zero probability path, is the language $\mathcal{L}(\mathcal{M})$ of the HMM. The ith observation of a trace is denoted as $\tau^{(i)}$. We define $Pr(\tau \mid \pi) = 1$ if $obs(\pi) = \tau$ and zero otherwise. The probability of a trace τ is $\sum_{\pi \in \Pi^{\mathcal{M}}} Pr(\pi) \cdot Pr(\tau \mid \pi)$. Finally, the conditional probability on a trace $\tau \in \mathcal{L}(\mathcal{M})$ is defined using Bayes' rule $Pr(\pi \mid \tau) := {}^{Pr(\tau|\pi) \cdot Pr(\pi)}/_{Pr(\tau)}$.

Example 1. We consider the HMM from Fig. 3a. The conditional probability $Pr(q_c, q_i, q_i \mid dry, icy, icy)$ is ${}^{Pr(dry, icy, icy \mid q_c, q_i, q_i) \cdot Pr(q_c, q_i, q_i)}/_{Pr(dry, icy, icy)}$. $Pr(dry, icy, icy \mid q_c, q_i, q_i)$ is 1, and $Pr(q_c, q_i, q_i)$ is $^9/_{40}$. The sum of all paths which observe dry, icy, icy is $^{11}/_{20}$. Thus, the total probability of q_c, q_i, q_i given dry, icy, icy is $^9/_{22}$.

2.2 Formal Problem Statement

Definition 3 (Monitor). *A DFA $\mathcal{A}$ is a monitor for HMM $\mathcal{M}$ if the alphabet for $\mathcal{A}$ coincides with the observations in $\mathcal{M}$.*

Monitors should accept unsafe traces, which we define via their risk [27]:

Definition 4 (Trace risk, safe/unsafe traces). *Given HMM $\mathcal{M}$, the* risk *of $\tau \in \mathcal{L}(\mathcal{M})$ is:*

$$R(\tau) := \sum_{\pi \in \Pi^{\mathcal{M}}_{|\tau|}} Pr(\pi \mid \tau) \cdot r(\pi_\downarrow).$$

Let $\lambda_s \leq \lambda_u \in \mathbb{R}$ be the safe threshold *and an* unsafe threshold*. A trace $\tau \in \mathcal{L}(\mathcal{M})$ with $R(\tau) > \lambda_u$ is* unsafe*, while τ is* safe *if $R(\tau) < \lambda_s$.*

[2] We use determinstic observation functions for concise definitions. Stochastic observation functions can be expressed via a blowup of the HMM, see e.g. [27].

We deliberately do not require $\lambda_s = \lambda_u$. By picking $\lambda_s < \lambda_u$, some traces are neither safe nor unsafe, also called inconclusive. We write $\mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$ (and $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u}$) for the set of safe (and unsafe) traces of length at most h.

Example 2. We consider the HMM from Fig. 3a, with the risk function assigning 1 to q_c and 0 to all other states. Taking the trace *dry*, *icy*, *icy*, there are three paths which could generate this trace. Two paths end in q_c, one ends in q_i. The paths ending in q_c have a conditional probability of $13/22$. Since only these paths have a non-zero risk, the risk of the trace is $13/22 \cdot r(q_c) = 13/22$.

Definition 5 (Missed/False alarms). *Given a monitor $\mathcal{A}$ for HMM $\mathcal{M}$, a horizon h, and thresholds $\lambda_s \leq \lambda_u \in \mathbb{R}$, the set of* missed alarms *is* $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \setminus \mathcal{L}(\mathcal{A})$. *The set of* false alarms *is* $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s} \cap \mathcal{L}(\mathcal{A})$.

Definition 6 (Correct monitor). *Given thresholds λ_s, λ_u and horizon h, a monitor $\mathcal{A}$ for $\mathcal{M}$ is* correct *if* $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset = \mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A})$.

A correct monitor raises an alarm for all unsafe traces and for no safe trace, i.e., missed alarms are false negatives, while false alarms are false positives.

Corollary 1. *A monitor $\mathcal{A}$ is correct iff* $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \subseteq \mathcal{L}(\mathcal{A}) \subseteq \Sigma^* \setminus \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$.

Problem statements. Given HMM $\mathcal{M}$, thresholds λ_s, λ_u and horizon h:

1. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there no-missed-alarms, i.e., is $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset$?
2. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there no-false-alarms, i.e., is $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A}) = \emptyset$?
3. Find a correct monitor $\mathcal{A}$ for $\mathcal{M}$ w.r.t. λ_s, λ_u and h.

Problems 1 and 2 together allow checking whether a monitor is correct. Furthermore, a correct monitor must exist, as $\mathbb{U}^{\leq h}$ is finite and thus regular.

Example 3. We discuss the example from Fig. 3 using the correct monitor, $\mathcal{A}$. Given the risk function assigning 1 to q_c and 0 to all other states, the traces $\tau_1 =$ *dry*, *icy*, *icy* and $\tau_2 =$ *dry*, *icy* have risks $13/22, 1/10$ respectively. If λ_s is $1/4$ and the horizon h is 3, τ_2 is the trace with maximum risk not accepted by the monitor. Given that its risk is below λ_s, the monitor does not have any missed alarms. Similarly, monitor $\mathcal{A}$ does not have any false alarms for $\lambda_u = 1/4$. Thus, $\mathcal{A}$ is a correct monitor for $\mathcal{M}$ with h, λ_s, and λ_u.

3 Monitor Verification

We present our approach to the monitor correctness problem, which reduces checking the existence of missed alarms to policy synthesis on colored MDPs, defined below. The latter problem is well studied. We first formalize the policy synthesis problem and then discuss the transformation in multiple steps. Here, we focus on showing that there are no missed alarms of exactly the length of the horizon (an adaption of Problem 1). At the end of the section, we generalize our construction to finding false alarms and to traces of length *at most* the horizon.

3.1 Relating Missed Alarms with Color-Consistent Policies

A policy for MDP $\mathcal{M}$ is a function $\sigma\colon \Pi^{\mathcal{M}}_{fin} \to Act$, which selects actions for every finite path. An MDP and a policy induce an MC. Policy synthesis for a MDP for a property ϕ entails finding a policy for an MDP such that the induced MC entails ϕ.

Luko: Should we mention memoryless schedulers? We use σ as $S \to A$

Colored (Labelled) MDPs are an extension to MDPs that allow expressing dependencies between states that policies must adhere to. The following definition simplifies the notion of colored MDPs from [**empty citation**], and suffices for our needs:

Definition 7 (Colored MDP). *Given an MDP $\mathcal{M}$ with states S, a* colored MDP *is a tuple $\mathcal{M}^C = (\mathcal{M}, C, c)$, where C is a set of* colors*, and $c\colon S \to C$ maps states to colors.*

Luko: Make sure we always use $\mathcal{M}^C$ for colored MDPs.

Definition 8 (Color consistent). *A memoryless scheduler σ for a colored MDP $\mathcal{M}^C$ is* color consistent[3] *if for states s, s', $c(s) = c(s')$ implies $\sigma(s) = \sigma(s')$.*

The set of all color-consistent schedulers is denoted Σ_c. Policy synthesis for colored MDPs asks to find a color-consistent scheduler such that the reachability probability to a set of target states is above a certain threshold. Policy synthesis for colored MDPs is NP-hard[**empty citation**].

Theorem 1 (Main theorem). *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, safe threshold λ_s, horizon h, and risk r, there is a colored MDP $\mathcal{M}^C$ with target states T, and threshold λ s.t.*

$$\exists \sigma \in \Sigma_c.\ Pr_{\sigma}^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Our proof, outlined in this section, is constructive and we show that we can use the construction to find a $\tau \in \mathsf{mA}(\mathcal{A})$, whenever such a τ exists.

Outline of the proof. The proof is a direct consequence of Lemmas 1 to 3 below. Before we discuss these lemmas, we observe that on the left-hand side of Theorems 1, the monitor, horizon, observations, and risk do not occur, they must be encoded into the colored MDP. Furthermore, while missed alarms are defined using conditional probabilities, the policy synthesis problem is over reachability probabilities. We describe our transformation in several steps. In Section 3.2 we encode the monitor into the HMM and transform the HMM to both include the horizon and the risk. In Section 3.3 we resolve the conditioning on traces, while replacing observations from the HMM.

3.2 The (Acyclic) Conditional Trace Probability Problem

First, we show how asking for a missed alarm can be rephrased into the conceptually simpler *conditional trace probability* (CTP) problem. We will further simplify the problem by incorporating both the horizon and risk such that we are left with a problem purely on acyclic HMMs.

[3] Colored MDPs with color-consistent policies coincide with memoryless policies for partially observable MDPs. However, POMDPs often consider history-dependent (belief-based) policies. We use *colored MDPs* to avoid any confusion.

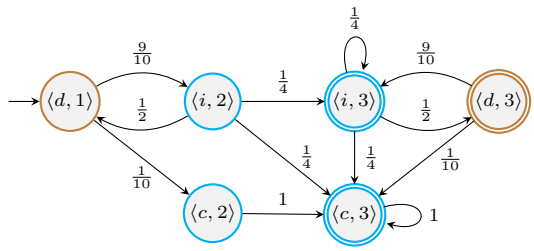


Fig. 4: The central example HMM and faulty monitor $\mathcal{B}$ transformed into the CTP problem. States are named with their HMM state, $\{d, i, c\}$ and their monitor state, $\{1, 2, 3\}$. States in the alarm states set F have been marked accepting.

CTP problem We take (a mild variation of) a standard product construction [12] between HMM and the DFA. We define as *alarm states* F as those states which correspond to non accepting states in the DFA. This is equivalent to taking a product with the complement of the monitor.

Example 4. Figure 4 shows the product of the HMM and monitor $\mathcal{B}$ from Figure 3. Starting in the initial states d and 1, the HMM can transition to the i state with probability $9/10$. This is an *icy* state, and thus the monitor takes the *icy* transition to the 2 state. This results in the new product state of i and 2, with a transition from $\langle d, 1\rangle$ to $\langle i, 2\rangle$ with probability $9/10$. The alarm states are any product states $\langle _, 3\rangle$.

Definition 9 (HMM product). *Given a HMM* $\mathcal{M} = (S, \iota^{\mathcal{M}}, \mathbf{P}, Z, obs, r)$ *and monitor* $\mathcal{A} = (Q, \Sigma, \delta, \iota^{\mathcal{A}}, F')$*, the* product HMM $\langle \mathcal{M}_{\times\mathcal{A}}, F\rangle$ *is the HMM* $\mathcal{M}_{\times\mathcal{A}} = (S \times Q, \langle \iota^{\mathcal{M}}, \iota^{\mathcal{A}}\rangle, \mathbf{P}', obs', r')$ *with* $obs'(\langle s, q\rangle) = obs(s)$*,* $r'(\langle s, q\rangle) = r(s)$*,*

$$\mathbf{P}'(\langle s, q\rangle)(\langle s', q'\rangle) = \begin{cases} \mathbf{P}(s, s') & \text{if } \delta(q, obs(s')) = q', \\ 0 & \text{otherwise,} \end{cases}$$

and the alarm states $F = S \times F'$*.*

In the product, we can find a trace τ whose conditional trace probability exceeds a threshold iff τ is a missing alarm. We state the decision problem that needs to be solved: It is key to our computational complexity analysis in Seciton 5.

Definition 10 (CTP Decision Problem). *Given HMM* $\mathcal{M}$ *with states* S *and risk* r*, (unary encoded) horizon* h*, alarm states* $F \subseteq S$*, and threshold* $\lambda_s \in \mathbb{R}_{\geq 0}$*,*

$$\exists \tau \in \mathcal{L}(\mathcal{M}).\ \sum\nolimits_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in F} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}) \geq \lambda_s.$$

We denote the set of witnesses τ to a CTP decision problem as $\mathsf{CTP}(\mathcal{M}, h, F, \lambda_s)$. The following lemma states the correctness of the transformation and follows directly from the definition of missing alarms and the product with the complement.

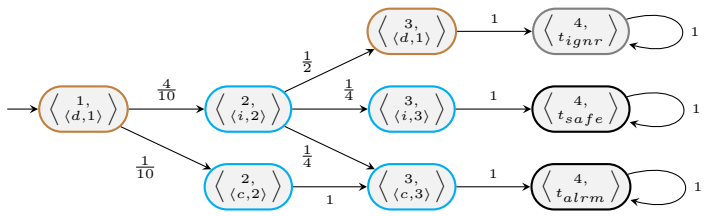


Fig. 5: ACTP, brown and cyan are dry and icy observations. Black is the new z_end observation, and gray is the new z_ignore observation. All states are named with the step i, model state s and monitor state j as $\langle i, \langle s, j\rangle\rangle$.

Lemma 1. *Using the notation from Theorems 1, Definition 9 and Definition 10:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}_{\times\overline{\mathcal{A}}}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

We provide a proof sketch in App. A.

ACTP problem We further simplify the problem by unrolling the model along the horizon. We also eliminate the risk function.

Example 5. Our unrolling applied to the HMM from Fig. 4 can be seen in Fig. 5. We unroll the product for horizon three. Thus, the initial state becomes the tuple of step 1 and the initial state from the CTP HMM, $\langle d, 1\rangle$. This state $\langle 1, \langle d, 1\rangle\rangle$ transitions to, e.g., $\langle 2, \langle c, 2\rangle\rangle$. Consider state $\langle 3, \langle i, 3\rangle\rangle$, it is at the horizon and its corresponding CTP state is in the alarm states F, the monitor raises an alarm. The normalized risk for this state is 0, since $r(\langle d, 1\rangle) = 0$. Thus, $\langle 3, \langle i, 3\rangle\rangle$ transitions with probability 1 to the safe state, $\langle 4, t_{safe}\rangle$, and probability 0 to the alarm state, $\langle 4, t_{alrm}\rangle$. If we consider $\langle 3, \langle d, 1\rangle\rangle$, whose CTP state is not in F, but is at the horizon, it transitions to the ignore state, $\langle 4, t_{ignr}\rangle$.

Definition 11 (Unrolling with risk). *The (risk-)unrolled HMM $\mathcal{M}_{\blacktriangleright h}$ of a HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ with horizon h and alarm states $F \subseteq S$ is the HMM $\mathcal{M}_{\blacktriangleright h} = (\{1, \ldots, h\} \times S \cup \{\langle h+1, t_{alrm}\rangle, \langle h+1, t_{safe}\rangle, \langle h+1, t_{ignr}\rangle\}, \langle 1, \iota\rangle, \mathbf{P}', Z \cup \{z_{end}, z_{ignore}\}, obs')$, with the new transition function:*

$$\forall_{i \in \{1,\ldots,h-1\}} \quad \mathbf{P}'(\langle i, s\rangle)(\langle i+1, s'\rangle) = \mathbf{P}(s)(s')$$

$$\mathbf{P}'(\langle h, s\rangle)(\langle h+1, t\rangle) = \begin{cases} \frac{r(s)}{\max_{s \in S} r(s)} & \textit{if } s \in F \wedge t = t_{alrm}, \\ 1 - \frac{r(s)}{\max_{s \in S} r(s)} & \textit{if } s \in F \wedge t = t_{safe}, \\ 1 & \textit{if } s \notin F \wedge t = t_{ignr}, \\ 0 & \textit{otherwise} \end{cases}$$

$$\mathbf{P}'(\langle h+1, t\rangle)(\langle h+1, t\rangle) = 1$$

Where the observation function obs' for states $\langle i, s\rangle$ is given by $obs(s)$, the t_{ignr} has the z_{ignore} observation, and t_{alrm} and t_{safe} have the z_{end} observation.

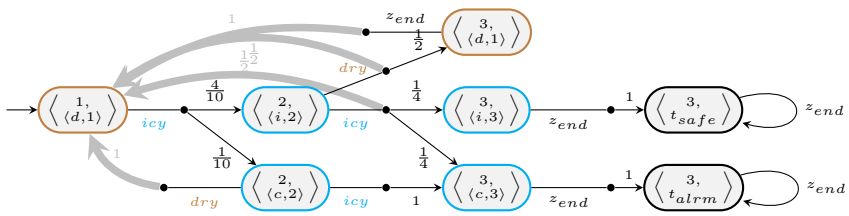


Fig. 6: The central example transformed into an mdp to be used as the model in the MDP policy synthesis problem. For conciseness, all unreachable states are removed. Also any actions which return for all states in the same step are removed.

For the first h steps, the unrolling is standard. At the horizon we transition to the three dedicated states t_{alrm}, t_{safe} and t_{ignr}[4] according to the risk.

Lemma 2. *Given an HMM $\mathcal{M}$, horizon h, alarm states F, and threshold λ_s, there exists a $\lambda \in (0, 1]$ such that:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi \cdot t_{alrm} \mid \tau \cdot z_{\mathsf{end}}) \geq \lambda,$$

where z_{end} is the final observation and t_{alrm} is the alarm state.

SJ: I wonder whether we should simplify notation on the right hand side. I am not too fond of the z and s_a symbols.

We provide a proof in App. A.

3.3 Reduction to Consistent Policy Synthesis

For Lemma 2, we must find a trace such that a conditional reachability probability exceeds a threshold. We reformulate this into a policy synthesis problem in MDPs. The transformation combines two ideas: First, in every state, the policy can select the next observation, loosely inspired by [10]. Second, we reformulate conditional reachability probabilities by (standard) regular reachability probabilities, using a construction in [27, 13].

Example 6. We transform the HMM from Fig. 5 to the Colored MDP from Fig. 6. Consider $\langle 2, \langle i, 2\rangle\rangle$ in original the HMM. The next observation is either *dry* or *icy*, which corresponds to two new actions for $\langle 2, \langle i, 2\rangle\rangle$ in the colored MDP. The *dry* action transitions to the ensuing states with the *dry* observation. The remaining probability of $1/2$ for the *dry* action is directed to the initial state. This is similarly applied to the *icy* action. State $\langle 2, \langle c, 2\rangle\rangle$ does not reach any *dry* states, we still add a *dry* action redirecting to the initial state, since we want all states with the same step value to have the same actions. State $\langle 3, \langle d, 1\rangle\rangle$ in

[4] We add the $\langle h+1, t_{ignr}\rangle$ state, instead of redirecting all traces we don't care about to the $\langle h+1, t_{safe}\rangle$ state, to more easily modify the transformation for the no-false-alarms problem (See Sec. 3.4).

the HMM only has a transition to an z_{ignore} state. We fully remove the z_{ignore} state and the z_{ignore} observation in the MDP. Finally, we make sure all states with step 3 have the same actions, and thus add the z_{end} action redirecting to the initial state.

Definition 12. *Given an acyclic HMM $\mathcal{M}_{\blacktriangleright h} = (S, \iota, \mathbf{P}, Z, obs, r)$ as in Definition 11, we define the colored MDP $\mathcal{M}_{\gtrdot h} = ((S \setminus \{t_{ignr}\}, \iota, Act, \mathbf{P}'), C, c)$ with $Act = Z \setminus \{z_{ignore}\}$,*

$$\mathbf{P}'(\langle i, s\rangle, z)(\langle j, q\rangle) := \begin{cases} \mathbf{P}'(\langle i, s\rangle)(\langle j, q\rangle) & \text{if } obs(\langle j, q\rangle) = z, \\ \sum_{(\langle k,q'\rangle)\in S' \mid obs(\langle k,q'\rangle)\neq z} \mathbf{P}'(\langle i, s\rangle)(\langle k, q'\rangle) & \text{if } (\langle j, q\rangle) = \iota, \\ 0 & \text{otherwise} \end{cases}$$

$C = \{1, \ldots, h\}$ and c is given such that $c(\langle i, s\rangle) = i$.

Thus, we transition normally to a state if the action and observation of the target state are equal, otherwise we set the transition probability to zero. All the remaining probability mass is redirected towards the initial state.

Remark 1. All states gain all actions in $\mathbf{P}'$, we can later prune any actions for all states S^C of a color C when these actions only redirect to the initial state from S^C.

The above construction allows for conditioning on a trace τ by constructing a policy σ that selects the ith observation of τ in the state with step i.

Definition 13 (Trace consistent policy). *Given an MDP $\mathcal{M}_{\gtrdot h}$ as in Def. 12. Together with a trace $\tau \in Z^\star$. A* trace consistent policy *is of the form*

$$\sigma_\tau(\langle i, s\rangle) = \begin{cases} \tau^{(i)} & \text{if } i \leq |\tau|, \\ end & \text{otherwise.} \end{cases}$$

Using the coloring as described in Def. 12, the trace consistent policies coincide with the color consistent policies. Finding a missed-alarm traces now reduces to solving the following color consistent policy synthesis problem:

$$\exists \sigma \in \Sigma_c . Pr^\sigma(\Diamond \langle h+1, t_{alrm}\rangle) > \lambda$$

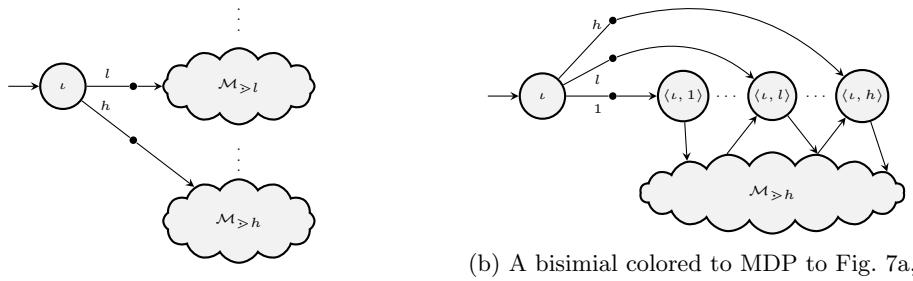
Which is solved using PAYNT [6].

Lemma 3. *Given an HMM $\mathcal{M}$, horizon h, alarm states F, target state T and threshold λ:*

$$\exists \sigma \in \Sigma_c . Pr_\sigma^{\mathcal{M}_{\gtrdot h}}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}_{\blacktriangleright h}}} Pr^{\mathcal{M}_{\blacktriangleright h}}(\pi \cdot t_{alrm} \mid \tau \cdot z_{end}) \geq \lambda,$$

We provide a proof sketch in Appendix A.

Corollary 2. *There exists a map $t(\sigma_\tau) = \tau$ which, given a trace consistent policy σ_τ, finds its associated trace τ.*



(a) A colored MDP combining the colored MDPs from Def. 12 for horizons up to h, $\{\mathcal{M}_{\geqslant l} \mid l \in \{1, \ldots, h\}\}$, into one MDP.

(b) A bisimial colored to MDP to Fig. 7a, containing only $\mathcal{M}_{\geqslant h}$. States $\langle \iota, 1 \rangle$ through $\langle \iota, h \rangle$ from $\mathcal{M}_{\geqslant h}$ are shown separate in order to show their relation to ι.

Fig. 7: Transformation steps needed for Thm. 2.

3.4 Adapting to No-False-Alarms and Smaller Traces

Traces of Length up to the Horizon The approach for Thm. 1 only works for traces of length exactly the horizon. We generalize this theorem to traces of length at most the horizon.

Luko: Make sure this still matches Thm. 1.

Theorem 2. *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, safe threshold λ_s, horizon h, and risk r, there is a colored MDP $\mathcal{M}^C$ with target states T, and threshold λ s.t.*

$$\exists \sigma \in \Sigma_c.\ Pr_{\sigma}^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

The main insight for this theorem is show in Fig. 7. We combine the colored MDPs given by Thm. 1 for horizons 1 to h into one colored MDP, such that a policy starts by choosing which length trace to use (Fig. 7a). Using bisimulation minimization, we can transform this combined MDP back into just $\mathcal{M}_{\geqslant h}$ with a small addition (Fig. 7b). We give a more detailed explanation in Appendix A.

Luko: Reference Prob. 2 somewhere here.

No-False-Alarms Problem We modify the transformation from Thm. 2 such that it solves the no-false-alarms problem. The no-false-alarms problem differs in two ways from the no-missed-alarms problem. We are finding a trace *accepted by* the monitor, and we find a trace whose risk is *below* the *unsafe* threshold.

Theorem 3. *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, safe threshold λ_s, horizon h, and risk r, there is a colored MDP $\mathcal{M}^C$ with target states T, and threshold λ s.t.*

$$\exists \sigma \in \Sigma_c.\ Pr_{\sigma}^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A}).$$

We provide a proof sketch for this theorem in Appendix A, however, we highlight the main ideas here. In order to find a trace accepted by the monitor, we no longer take the complement of the monitor while transforming to CTP. To find a safe trace we do reachability on t_{safe} instead of t_{alrm} while taking as a

threshold $1 - \lambda$. Thus, we find a trace whose probability of being safe is above a threshold[5].

4 Learning Correct Monitors

We describe how to *learn* correct monitors (Prob. 3), by combining automata learning with a *Minimally Adequate Teacher* (MAT, [7]) and monitor verification.

MAT framework. We briefly recap the MAT framework, for details see [38]. A minimally adequate teacher answers two types of questions: A *membership query* (MQ), which in our setup means *should a trace be accepted by the monitor?*, and an *equivalence query* (EQ), *is this monitor correct?* Furthermore, if the answer to an equivalence query is negative, we must provide a counterexample that witnesses why the monitor is not correct. Various algorithms implementing the MAT framework for DFA learning exist. For the purpose of this paper, we use the $L^\star$ algorithm [7] to learn a monitor. The learner asks MQs to the teacher until a *hypothesis monitor* can be constructed which is consistent with the MQs. Once such a hypothesis is constructed, its correctness is verified using an EQ.

Verification as a MAT. To learn a monitor $\mathcal{A}$, we provide the HMM $\mathcal{M}$, a risk function r, a horizon h, a learning threshold λ_l, and the safe and unsafe thresholds λ_s and λ_u. The additional learning threshold λ_l is used to define an MQ whenever $\lambda_s \neq \lambda_u$: In particular, for MQs, each trace must be unambiguously safe or unsafe: the MAT framework does not allow for flagging certain traces as *don't care*, while traces with a risk between λ_s and λ_u can be considered don't care in our setting. Likewise, the MQ must also be defined for traces $\tau \notin \mathcal{L}(\mathcal{M})$ or traces longer than the horizon. We thus adapt the notion of safe traces from Def. 4.

Definition 14. *Given any trace $\tau \in Z^\star$ and a horizon h, membership query MQ_{λ_l} is a function such that $\mathrm{MQ}_{\lambda_l}(\tau)$ is unsafe iff $\tau \in \mathcal{L}(\mathcal{M})$[6] and $\tau \in \mathbb{U}_{\lambda_l}^{\leq h}$.*

Such a function for MQ_{λ_l} can be defined by keeping track of the probability of being in each state after every observation from the trace *or* by model checking the induced Markov chain that reflects the trace-consistent policy in Seciton 3 [30, 27]. For EQs, we simply use the notion of correctness from Def. 6.

Definition 15. *Given a HMM $\mathcal{M}$, and a monitor $\mathcal{A}$, an $\mathrm{EQ}_{\lambda_s,\lambda_u}$ is a function $\mathrm{EQ}_{\lambda_s,\lambda_u}(\mathcal{A}) \in \{\top\} \cup Z^\star$. Such that, $\mathrm{EQ}^{\mathcal{M}}_{\lambda_s,\lambda_u}(\mathcal{A})$ holds if $\mathcal{A}$ is correct for $\mathcal{M}$ with λ_s, and λ_u (in the sense of Def. 6), and $\mathrm{EQ}^{\mathcal{M}}_{\lambda_s,\lambda_u}(\mathcal{A})$ returns the missed alarm or false alarm trace for an incorrect $\mathcal{A}$.*

The EQ requires checking both for no-missing-alarms, and for no-false-alarms. Each check follows the steps as described in Sec. 3.

Lemma 4. *Given a MAT with a $\mathrm{EQ}_{\lambda_s,\lambda_u}$ and a MQ_{λ_l}, a monitor learned with $L^\star$ is correct as long as $\lambda_s \leq \lambda_l \leq \lambda_u$.*

[5] We cannot find a trace whose risk is below a threshold since minimizing reachability of t_{alrm} will try to choose a path that never takes a z_{end} action.

[6] Defining traces $\tau \notin \mathcal{L}(\mathcal{M})$ as safe is an arbitrary design decision.

Proof. If, while learning a monitor $\mathcal{A}$ a trace τ is deemed *safe* by MQ_{λ_l} it cannot be given as a counterexample by $\mathrm{EQ}_{\lambda_s,\lambda_u}$ on $\mathcal{A}$, since $\lambda_s \leq \lambda_l \leq \lambda_u$, and $\tau \notin \mathcal{A}$ by $\mathrm{L}^\star$. Similarly, MQ_{λ_l} and $\mathrm{EQ}_{\lambda_s,\lambda_u}$ also agree on *unsafe* traces. Now, by correctness of $\mathrm{L}^\star$, a learned monitor $\mathcal{A}$ has to be correct according to $\mathrm{EQ}_{\lambda_s,\lambda_u}$, and thus correct in the sense of Def. 6.

When $\lambda_s < \lambda_u$, the EQ has an inconclusive area given by the interval (λ_s, λ_u). This means that our EQ does not check for equivalence, but simply accepts any correct monitor. We investigate the effect of this inconclusive area in Sec. 6.

Conformance queries. An alternative to the EQ in Def. 15 is a conformance query [20]. It tests a monitor by sampling traces from the HMM and checking if the MQ and the monitor agree. If all the monitor and the MQ don't agree on a trace, it is given as a counterexample. In our approach we use a hybrid of the two EQs. Monitors produced early in the learning process often contain many missed alarms and false alarms. Verification can find them, however, applying the transformation from Sec. 3 has a constant cost. Conformance queries can often find a counterexample faster if they have a high probability of occurring.

5 Computational Complexity

This section discusses the hardness of monitor verification (Thm. 4) and the inapproximability of a related optimization problem (Lem. 6).

Theorem 4. Is a monitor correct? *(w. unary coded horizon) is coNP-complete.*

In fact, we study the dual to this problem, i.e., checking the existence of a counterexample. We call this problem *monitor co-verification*. For monitor co-verification, *membership* in NP follows as false alarms or a missed alarm (of length up to horizon, i.e., polynomially sized) are witnesses. Verifying whether a trace is a false or missed alarm can be done in polynomial time, by checking whether the automaton accepts it and computing the trace risk (see Sec. 4).

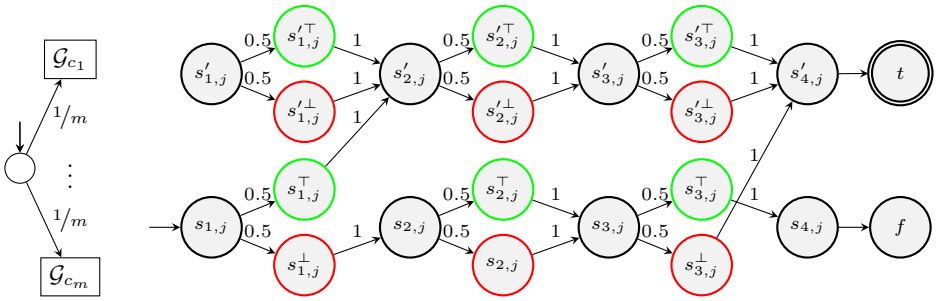
To establish NP-hardness, we consider the CTP problem from Definition 10. As a solution to the monitor co-verification problem solves the CTP problem (using a trivial monitor), this implies NP-hardness of the former problem.

Lemma 5. *The CTP Decision Problem is (strongly) NP-hard.*

The proof features a reduction from SAT, the problem of satisfiability of a propositional formula. We illustrate the reduction, details are in App. B.

We construct HMM $\mathcal{M}_\varphi$ from CNF φ over variables X such that there is a trace with risk 1 iff there is a satisfying assignment to φ. In particular, that trace exists iff all thre is a trace τ s.t. all corresponding paths reach some state t. The traces are of the form $\#\#\cdot\{\bot,\top\}\cdot\#\cdot\{\bot,\top\}\cdots\#$: Trace $\#\#\cdot\alpha(x_0)\cdot\#\cdot\alpha(x_1)\cdots\#$ represents assignment $\alpha\colon X \to \{\bot,\top\}$. We now construct $\mathcal{M}_\varphi$ such that any trace that ensures reaching t reflects a satisfying assignment. We create gadgets for every clause. The gadgets are connected as in Figure 8a: That is, to ensure reaching t along every path, we must reach t in every gadget. The gadget $\mathcal{G}_j$

SJ: make sure that corresponding paths are the right formal notion



(a) HMM. (b) Gadget $\mathcal{G}_c$ demonstrated for $c = x_1 \vee \neg x_3$

Fig. 8: Illustrations for Lemma 5, with m clauses and variables x_1, x_2, x_3.

intuitively 'evaluates' c_j with respect to an assignment, as exemplified in Fig. 8b. A path (or its trace) through $\mathcal{G}_j$ 'reads' variable x_i in state $s_{i,j}$ and transitions to $s^{\top}_{i,j}$ or to $s^{\bot}_{i,j}$. The states are labelled $\#, \top, \bot$, respectively. However, for a trace where $\alpha(x_i) = \top$, only the former path corresponds to the trace (and symmetrically for $\alpha(x_i) = \bot$. That path reaches state t iff the assignment satisfies at least one literal in the clause.

We now show that the construction above suffices to show that it is hard to approximate the maximal risk that a monitor admits.

Definition 16 (CTP Optimization Problem). *Given an HMM $\mathcal{M}$ with states in S, a unary encoded horizon h, a set of alarm states $F \subseteq S$:*

$$\max_{\tau \in \mathcal{L}(\mathcal{M})} \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in F} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}).$$

Lemma 6. *The CTP Optimization Problem is APX-hard.*

This follows from a strict reduction from MAX-3SAT, which is an inapproximable and APX-hard problem [22]. The construction coincides with the reduction in Lemma 5 by observing that the conditional probability to reach a t state is given by $1/m$ times the number of satisfied clauses, i.e., we can compute the maximal number of satisfied clauses in φ on the HMM $\mathcal{M}_{\varphi}$.

6 Experiments

We empirically evaluate the monitor verification (Sec. 3) and monitor learning (Sec. 4) using our prototype implementation called ToVer. Code, benchmarks, and logs will be publicly available via the artifact evaluation.

Setup. The ToVer tool is implemented in Python and C++ on top of the model checker Storm [24] for data structures and for the MQs in Sec. 4 [27]. We use PAYNT [6] to verify colored MDPs (Def. 7), using exact arithmetic to avoid numerical problems on these types of benchmarks [21]. The learner uses the

		Benchmark								ToVer				
		h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\gg h}\rvert$	λ^{found}
AirportA-3	A-0	10	MA	45	113	18	23	414	10^{12}	1.82	$\leq 1s$	1.64	328	0.30
AirportA-3	A-2	10	FA	45	113	18	23	414	10^{13}	6.00	$\leq 1s$	5.82	328	0.19
AirportA-3	A-8	10	MA	45	113	18	81	1458	10^{12}	900.48	$\leq 1s$	900.08	✓	✓
AirportA-3	A-10	10	FA	45	113	18	81	1458	10^{13}	1070.69	$\leq 1s$	1070.27	✓	✓
Evade	E-20	9	MA	385	1473	325	288	93600	10^{14}	79.85	70.47	9.24	✓	✓
Evade	E-22	9	FA	385	1473	325	288	93600	10^{14}	79.30	70.28	9.01	✓	✓
Hidden-Incen.	H-8	10	MA	397	1649	100	257	25700	10^{18}	1233.95	10.10	1223.79	✓	✓
Hidden-Incen.	H-10	10	FA	397	1649	100	257	25700	10^{18}	4888.19	10.27	4877.91	✓	✓
Icy-Driving	I-34	3	FA	3	6	2	8	16	10^{0}	$\leq 1s$	$\leq 1s$	$\leq 1s$	✓	✓
Icy-Driving	I-14	25	FA	3	6	2	2	4	10^{11}	1393.58	$\leq 1s$	1393.57	✓	✓

Table 1: Subset of verification results found in Appendix C. The columns give the family name, an ID, horizon, and whether we check for missed alarms or false alarms. We give the size of the HMM (states, transitions), the number of observations, the size of the DFA (states, transitions), and the size of the language after pruning unreachable states. Furthermore, we list the run time for the complete verification procedure as well as the time spent on transforming the problem into a policy synthesis problem and the policy synthesis in PAYNT. Dashes in λ^{found} indicate no trace was found with a risk above (or below, for FA) the indicated threshold.

AAlpy framework [28]. All experiments are run on a single thread of an AMD Ryzen TRP 5965WX and with a memory limit of 15 GiB.

SJ: Maybe mention that for conformance queries, we sample traces in the HMM rather than using a test suite (test suites = the things that Loes does)?

Benchmarks. We take the benchmarks Aiport, Refuel, Evade, and Hidden-Incentive from [27]. We add Icy-Driving, a scaled-up version of the central example and SnL based on the game "Snakes and Ladders". While the benchmarks from the literature contain many observations, i.e., few states share an observation, the new benchmarks only have a few different observations. All benchmarks are scalable. The risk function is defined by a temporal property, e.g., the probability of reaching a bad state within a few steps.

Efficiency of Monitor Verification We investigate scalability along different dimensions and identify the bottlenecks of the method.

SJ: This probably requires a few more rows in the table

Setup. We obtain monitors from the ToVer learning experiments (below, with $\lambda_s = \lambda_u = 0.3$). Every version of the benchmark is run on the first (incorrect) monitor that passed a limited conformance check, an (incorrect) monitor obtained halfway through the learning process, and the final correct monitor. For each monitor we verify correctness w.r.t. the same λ_s, λ_u.

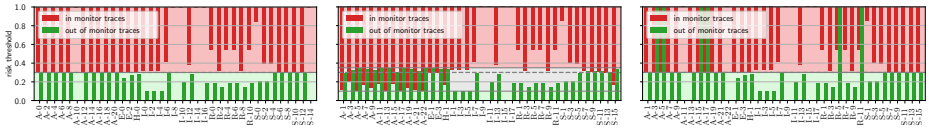
Luko: Fix table with new results and $\leq 1s$

Results. We present our results in Table 1, which is a subset of the 336 benchmarks shown in Appendix C. Generally, we observe that we verify HMMs with hundreds of states and up to hundred thousands of transitions, see benchmarks E-20/E-22. Benchmarks A-0/A-2 reflect verification w.r.t. almost trivial monitors, for which it is typically easy to find a counterexample, whereas A-8/A-10 reflect verification of the same HMM with respect to a larger correct monitor. Increasing

SJ: A dash looks like a bad thing, but maybe we should use a checkmark instead?

SJ: Can we say something about search space

Luko: TODO: add magnitude of product (for now min=8, avg=3180.43,



(a) ToVer, $\lambda_s = \lambda_l = \lambda_u$ (b) ToVer, $\lambda_s < \lambda_l < \lambda_u$ (c) Baseline

Fig. 9: Actual alarm and actual no-alarm thresholds from monitors learned with ToVer learning and baseline learning. The line between the green and gray area is λ_u, the line between the red and gray area is λ_s. The dotted line is λ_l. Missing bars reflect time-outs.

the horizon significantly increases the runtime, even for small models, e.g., I-14 compared to I-34. In all benchmarks, the runtime consists almost exclusively of creating the input to PAYNT (taking the product and creating the MDP) and in running PAYNT. The former runs in polynomial time in the size of the input (see Appendix C), whereas the latter uses various heuristics to avoid the exponential computation time. In the current implementation, except for the (comparably) large EVADE benchmarks, the fast majority is spent on PAYNT. The transformation is never the bottleneck.

Luko: Is it polynomial, because the plot is in log scale, so wouldn't that make it exponential. (Maybe I am just wrong)

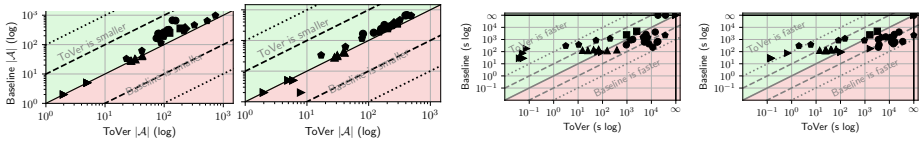
SJ: I mostly meant that its theoretical runtime is polynomial. Looking at the plot it is hard to see whether this grows exponentially or not. In particular, I would argue that it is polynomial in the size of the MDP, not in the size of the monitor or HMM.

Efficiency of Monitor Learning A key contribution of this paper is the ability to implement EQs for monitor learning. We consider the necessity of EQs, the size of the learned monitors, and the efficiency of learning them, both for $\lambda_s = \lambda_u$ and $\lambda_s \neq \lambda_u$.

Setup. We use the MAT framework from Sec. 4. Before every EQ, we run conformance checking (max. 100 samples using as threshold λ_s and 100 samples with λ_u, see Sec. 4). As hyper-parameters, we investigate (1) $\lambda_l = 0.3, \lambda_s = 0.35, \lambda_u = 0.1$ and (2) $\lambda_s = \lambda_l = \lambda_u = 0.3$ [7]. We compare against a baseline that does not use EQs, i.e., the baseline uses the MAT framework with only conformance checking (max. 100000 samples, different numbers of samples are tested in Appendix E).

Are the monitors correct? Using a proper EQ, we always learn correct monitors. We validate this experimentally *and* show that the baseline does not always yield correct monitors. Therefore, for any monitor we determine the unsafe trace with the lowest risk (*actual alarm threshold*, $\lambda_u^{\min}$) and the safe trace with the highest risk (*actual no-alarm threshold*, $\lambda_s^{\max}$). In a correct monitor, we have $\lambda_u^{\min} \geq \lambda_u$ and $\lambda_s^{\max} \leq \lambda_s$. Both Figures 9a and 9b and Fig. 9c show $\lambda_u^{\min}$ and $\lambda_s^{\max}$ for ToVer and baseline, respectively. Visually, a monitor is correct if its red bar never touches the green area, and the green bar never touches the red area. In 6 out of 38 benchmarks the baseline learns a monitor that misses alarms. No monitors had false alarms.

[7] We evaluate the role of λ_l on the correctness of monitors learned by the baseline in App. E.



(a) ToVer with $\lambda_s < \lambda_l < \lambda_u$ (b) ToVer with $\lambda_s = \lambda_l = \lambda_u$ (c) Runtime comparison, $\lambda_s < \lambda_l < \lambda_u$ (d) Runtime comparison, $\lambda_s = \lambda_l = \lambda_u$

Fig. 10: States in learned monitors and runtimes: ToVer vs baseline.

How big are the monitors? ToVer learns monitors with hundreds of states and tens of thousands transitions, see Figs. 10a and 10b (log-scale!) and Appendix D. For the literature on AAL, these are large automata[40, 2, 35]. Comparing the sizes of the monitors learned using ToVer and the baseline, monitors are smaller (up to 5 times, mostly at least 1.5 times smaller).

SJ: Why can they be smaller?

How fast do we learn the monitors? We compare the runtime of ToVer and baseline in the Figs. 10c and 10d (log-scale!). We remark that only ToVer is guaranteed to be correct. Neither of the two learning algorithms is clearly faster than the other, but ToVer has the potential to significantly accelerate the learning process, despite the high complexity. In particular, it tends to find counterexamples in the early stages much faster. This observation is independent of $\lambda_s = \lambda_u$ or $\lambda_s \neq \lambda_u$. In Appendix D, we detail where the time is spent. For most benchmarks, the EQ (in particular, PAYNT) is the bottleneck. However, for several EVADE benchmarks, most time is spent within L* code. We conjecture this happens as finding counterexamples is simple in these models.

SJ: How many EQs do we need to learn a monitor?

Luko: I added this to the table, but only for ToVer, I don't know this for the baseline.

The role of an inconclusive area. We compare between $\lambda_u < \lambda_l < \lambda_s$ and $\lambda_s = \lambda_l = \lambda_u$, i.e., with and without an inconclusive area. The baseline does not actively support such an inconclusive area. With an inconclusive area, more monitors are correct. The learner indeed finds monitors that are 2-5 times smaller (also compare Figures 10a and 10b). For benchmarks ICY-DRIVING, EVADE, and AIRPORT, this also translates to faster runtimes, sometimes by orders of magnitudes.

SJ: which table? Also make this explicit here

SJ: check numbers

7 Related Work

This work studies *monitoring based on stochastic systems* and combines *active learning* with *probabilistic verification*. We consider related work those directions.

SJ: TODO Check of werk van Stefan Kiefer et al over labelled MCs relevant is

Model-based monitoring for stochastic systems. Runtime verification is a wide field, see [18, 32, 23] for surveys. We review work on *model-based* runtime monitoring for stochastic systems. In particular, using state estimation on HMMs to decide whether to raise an alarm given a particular trace has been investigated in [33, 36, 39], extended to hybrid models [34], models with nondeterminism [27] and randomly timed models [10]. We use these techniques to answer membership

queries. The HMMs for runtime monitoring can be learned from a set of traces, see, e.g., [9, 8] and more recently [17], where they find the state risks at design time using model checking and use state estimation for runtime verification. Related to runtime monitoring is related to runtime enforcement, in particular shielding [15, 31, 19, 25]. Shielding is succesful in fully observable models but less studied in partial observable settings, in [**DBLP:conf/aaai/Carr0JT23**], shields are computed for qualitative properties. Finally, [1], a more general notion of correct monitors via linear time μ-calculus is investigated, while in [14] a notion of correct *predictors* is introduced. Both can be seen as generalizations of our definitions.

SJ: This sounds very close to what we do? Can we clarify the difference?

Learning monitors Learning monitors has been advocated in, e.g., [16, 29, 37, 41]. Closest to our setting is recent work in [26], which also uses state estimation for membership queries, but combines this with conformance queries and learns decision trees rather than automata. Crucially, by using conformance queries, the guarantees are significantly weaker, see also our experiments.

SJ: more? maybe also something by Anna Lukina?

SJ: Unsure whether we should say something about automata learning in general. Maybe better to have that only in section 4

Probabilistic verification The verification of our monitors applies model checking of conditional probabilities [13] to runtime verification, similar to [27, 10]. Most related is recent work in [10], where the models are CTMCs and the observation trace is uncertain itself. They also encounter a notion of trace-consistent policies, but instead of using synthesis, they overapproximate the verification by considering all policies. In contrast, our method is *complete*. Verification with partial observability as in our HMMs also occurs in the verification of partially observable MDPs [3], which can also be tackled using synthesis approaches [4]. Finally, considering MDPs as distribution transfomers [**empty citation**] gives a distinct semantics

SJ: add Peter van Rossum, TACAS 2006

SJ: stuff by Akshay S

SJ: can we say something useful here

8 Conclusion and Future Work

This paper presented a first approach to verification of monitors with respect to hidden Markov models. It embeds this verification procedure in an automata learning framework. The empirical evaluation is encouraging but also shows the limitations of the off-the-shelf frameworks. We see three avenues for future work: (1) Dedicated synthesis methods for conditional probabilities and the specific structure of our colored MDPs. (2) Automata learning for acyclic models and don't-care results. (3) Verification over unbounded (or very long) traces.

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A Proof Outlines for Seciton 3

Lemma 1. *Using the notation from Theorems 1, Definition 9 and Definition 10:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}_{\times\overline{\mathcal{A}}}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Proof Sketch. By the product construction, there exists a bijective mapping between paths in $M_{\times\overline{\mathcal{A}}}$ and paths in the monitor $\mathcal{A}$ and HMM $\mathcal{M}$. If the final state of a path is in F, the mapped path is not accepted by the monitor. The risk of a path in $M_{\times\overline{\mathcal{A}}}$ has the same risk as the mapped path in $\mathcal{M}$. Thus, if a trace is a witness for the CTP problem, it is a missed alarm for $\mathcal{M}$ and $\mathcal{A}$.

Lemma 2. *Given an HMM $\mathcal{M}$, horizon h, alarm states F, and threshold λ_s, there exists a $\lambda \in (0, 1]$ such that:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi \cdot t_{alrm} \mid \tau \cdot z_{\mathsf{end}}) \geq \lambda,$$

where z_{end} is the final observation and t_{alrm} is the alarm state.

SJ: I wonder whether we should simplify notation on the right hand side. I am not too fond of the z and s_a symbols.

Proof Sketch. We choose $\lambda = {}^{\lambda_s}/_{\max_{s \in S} r(s)}$. Both directions follow from applying Def. 10 and defining a bijective mapping f between paths in $\mathcal{M}_{\times\overline{\mathcal{A}}}$ (From the CTP problem) and $\mathcal{M}_{\blacktriangleright h}$ such that $Pr^{\mathcal{M}_{\times\overline{\mathcal{A}}}}(\pi) = Pr^{\mathcal{M}_{\blacktriangleright h}}(f(\pi))$. Now, for any path π of length h ending in a state in F, the transition of $f(\pi_{\downarrow})$ to t_{alrm} is equal to the normalized risk on $\pi_{\downarrow}$. From this we can conclude that risk for a trace in CTP is equal to the risk for a trace in ACTP, and thus that they have the same witnesses.

Lemma 3. *Given an HMM $\mathcal{M}$, horizon h, alarm states F, target state T and threshold λ:*

$$\exists \sigma \in \Sigma_c. Pr^{\mathcal{M}_{\geqslant h}}_{\sigma}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}_{\blacktriangleright h}}} Pr^{\mathcal{M}_{\blacktriangleright h}}(\pi \cdot t_{alrm} \mid \tau \cdot z_{end}) \geq \lambda,$$

Proof Sketch. We define a bijective map between finite paths π in $\mathcal{M}_{\blacktriangleright h}$ and a set X in the partition of the infinite paths in the induced MC by σ in $\mathcal{M}_{\geqslant h}$. The map is defined such that π followed by the infinite sequence of T is the postfix for every path in X. Using Def. 12, we can prove that the probability of π and X are equal. Now, using a bijective map t between trace consistent policies and traces, we can show that $Pr^{\mathcal{M}_{\geqslant h}}_{\sigma_\tau}(\Diamond T) = \sum_{\pi \in \Pi^{\mathcal{M}_{\blacktriangleright h}}} Pr^{\mathcal{M}_{\blacktriangleright h}}(\pi \cdot t_{alrm} \mid t(\sigma_\tau) \cdot z_{end})$. Thus, if there exists a trace consistent policy σ_τ above the threshold, $t(\sigma_\tau)$ is also above the threshold, and similarly using t^{-1} in the other direction.

Theorem 2. *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, safe threshold λ_s, horizon h, and risk r, there is a colored MDP $\mathcal{M}^C$ with target states T, and threshold λ s.t.*

$$\exists \sigma \in \Sigma_c.\ Pr^{\mathcal{M}^C}_{\sigma}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Proof. We modify the transformation from Thm. 1 in the policy synthesis step. We add a new initial state, ι', it gets a seperate coloring from all other states. This state contains an action for each possible length l of a trace up to the horizon. The actions now transition to the state $\langle l, \iota \rangle$.

$$\mathbf{P}'(\iota', l)(\langle l, \iota \rangle) = 1$$

We now show this transformation is correct.

In order to verify that there are no-missed-alarms for all $l < h$, we could use the transformation from Thm. 1 with the horizon equal to all $l < h$. This would neccesitate doing policy synthesis for h colored MDPs.

$$\exists_\tau \exists_{l \leq h} Pr^{\sigma_\tau}[\Diamond alarm] > \lambda_s$$

If any of these h policy synthesis problems can find a policy σ_τ, there exists a trace $\tau \in \mathsf{mA}_{\leq h}$.

We combine these h colored MDPs into one colored MDP in the following way. We add a new initial state, and give it h actions $\{1, \ldots, h\}$. Action $l \in \{1, \ldots, h\}$ points, with probability 1, to the initial state of colored MDP $\mathcal{M}_{\gtrdot l}$. This is equivalent to solving policy synthesis on the h individual MDPs.

Luko: Maybe note that the $\mathcal{M}_{\gtrdot l}$ only differed when they where in the CTP stage by the horizon, nothing else.

We now note that for any $l < l' \leq h$ the initial state $\langle 1, \iota^{CTP} \rangle$ of $\mathcal{M}_{\gtrdot l}$ is bisimilar to the state $\langle l' - l + 1, \iota^C TP \rangle$ in $\mathcal{M}_{\gtrdot l'}$. We now claim that the result of bisimulation minimization on this combined colored MDP is described by the transformation described at the start of the proof.

Theorem 3. *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, safe threshold λ_s, horizon h, and risk r, there is a colored MDP $\mathcal{M}^C$ with target states T, and threshold λ s.t.*

$$\exists \sigma \in \Sigma_c.\ Pr_\sigma^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{fA}^{\leq h}_{\mathcal{M}, \lambda_u}(\mathcal{A}).$$

Proof Sketch. The transformation from Sec. 3 is reused with the following differences. The complement of the monitor in Lem. 1 is no longer taken, and t_{safe} is used as the target state in Lem. 3. An outline of why the second step is correct is given below by showing the following.

$$\exists \sigma \in \Sigma_c. Pr_\sigma^{\mathcal{M}_{\gtrdot h}}(\Diamond t_{safe}) \geq 1 - \lambda$$
$$\Updownarrow$$
$$\exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}_{\blacktriangleright h}}} Pr^{\mathcal{M}_{\blacktriangleright h}}(\pi \cdot t_{alrm} \mid \tau \cdot z_{end}) \leq \lambda$$

B Construction for NP-hardness/APX-hardness

Consider a 3CNF formula $\varphi = \bigwedge c_1 \ldots c_m$ over variables X, $|X| = n$, with clause $c_j = \ell_j^1 \vee \ell_j^2 \vee \ell_j^3$ and each literal $\ell_j^i \in \{x, \neg x \mid x \in X\}$. We transform this into a CTP instance with $\lambda = 1$ and an acyclic HMM $\mathcal{M}_\varphi$ with observations $Z = \{\#, \bot, \top\}$. The only state with positive risk is a dedicated state t with $r(t) = 1$. The crux of the construction is that there is a trace with risk 1 iff there

SJ: make sure that corresponding paths are the right formal notion

is a satisfying assignment to φ. In the constructed HMM, there is a trace with risk 1 iff there is a trace where all corresponding paths end in state t.

Before we give a formal definition of $\mathcal{M}_\varphi$, we give some intuition. We represent assignments $\alpha\colon X \to \{\bot, \top\}$ by traces through the HMM $\#\#\cdot\alpha(x_0)\cdot\#\cdot\alpha(x_1)\cdots\#$. We create gadgets for every clause. The gadgets are connected as in Figure 8a: That is, intuitively, $\mathcal{M}_\varphi$ randomly selects a clause c_j with probability $1/m$ and transitions into gadget $\mathcal{G}_j$ defined below. Next, we show that the gadget reaches positive risk in every gadget only if the corresponding clause is satisfied by the assignment.

The gadget $\mathcal{G}_j$ intuitively 'evaluates' c_j with respect to an assignment α, by starting in $s_{1,j}$ with a gadget for this clause, as exemplified in Fig. 8b. We enter the gadget with trace $\# \cdot \alpha(x_0) \cdot \# \cdot \alpha(x_1) \cdots \#$, i.e., the first observation has been matched by the initial state. A path (or its trace) through $\mathcal{G}_j$ 'reads' variable x_i in state $s_{i,j}$ and transitions to $s^\top_{i,j}$ or to $s^\bot_{i,j}$. However, only one of these paths corresponds to the trace. Thus, in every gadget, there is only one path that corresponds to a given trace and so they can be used interchangeably. State $s_{i,j}$, $s^\top_{i,j}$ and $s^\bot_{i,j}$ have observations $\#, \top, \bot$, respectively. We note that a path reaches $s^\top_{i,j}$ if $\alpha(x_i) = \top$ and (analogously for $\bot$). From there onwards, if the literal x_i (or $\neg x_i$) occurs in clause c_j, we transition from $s^\top_{i,j}$ (or $s^\bot_{i,j}$, resp.) to $s'_{i+1,j}$, otherwise, we transition to $s_{i+1,j}$. A path ends in t only via a state $s'_{i,j}$, which is only possible if the path visits a state corresponding to a literal in the clause. Thus, (conditionally) reaching state $s'_{i,j}$ with positive conditional probability means that the partial assignment $\alpha(x_1), \ldots, \alpha(x_i)$ satisfies clause c_j, while reaching $s_{i,j}$ with positive conditional probability means that the clause is either unresolved or unsatisfied given the partial assignment.

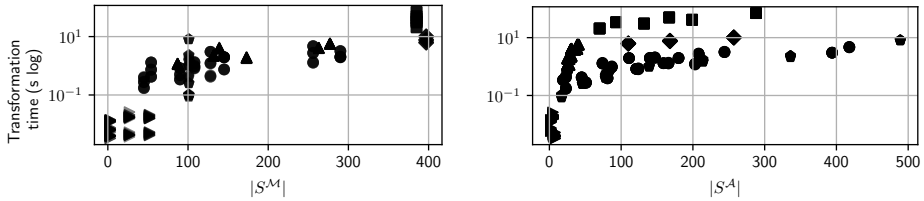
Formally, we construct the HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ and set $\lambda = 1$:

- $S = \{s_{i,j}, s^\bot_{i,j}, s^\top_{i,j}, s'_{i,j}, s'^\bot_{i,j}, s'^\top_{i,j} \mid i \in \{1, \ldots, n+1\}, j \in \{1, \ldots, m\}\} \cup \{s_\iota\}$.
- $\mathbf{P}$ is given such that for all $i \in \{1, \ldots, n\}, j \in \{1, \ldots, m\}$:
  - $\mathbf{P}(s_\iota, s_{1,j}) = \frac{1}{m}$,
  - $\mathbf{P}(s_{i,j}, s^\top_{i,j}) = \mathbf{P}(s_{i,j}, s^\bot_{i,j}) = \frac{1}{2} = \mathbf{P}(s'_{i,j}, s'^\top_{i,j}) = \mathbf{P}(s'_{i,j}, s'^\bot_{i,j})$,
  - $\mathbf{P}(s^\top_{i,j}, s'_{i+1,j}) = [x_i = \ell^k_j$ for some k$]$, $\mathbf{P}(s^\top_{i,j}, s_{i+1,j}) = [x_i \neq \ell^k_j$ for all k$]$,
  - $\mathbf{P}(s^\bot_{i,j}, s'_{i+1,j}) = [\neg x_i = \ell^k_j$ for some k$]$, $\mathbf{P}(s^\bot_{i,j}, s_{i+1,j}) = [\neg x_i \neq \ell^k_j$ for all k$]$
  - $\mathbf{P}(s'_{n+1,j}, t) = 1 = \mathbf{P}(s_{n+1,j}, f)$
- $Z = \{\top, \bot, \#\}$ and $obs(s^\top_{i,j}) = (s'^\top_{i,j}) = \top$, $obs(s^\bot_{i,j}) = (s'^\bot_{i,j}) = \bot$, $obs(s_{i,j}) = obs(s'_{i,j}) = \# = obs(s_\iota) = obs(t) = obs(f)$.
- $r(t) = 1$ and $r(s) = 0$ for all $s \in S \setminus \{t\}$.

The construction runs in polynomial time. The formal proof of its correctness follows the explanation above precisely.

C Results from ToVer Verification Experiments

We present the complete results of the monitor verification experiments. Figure 11 (log scale!) shows how transforming the HMM with the monitor into a colored



(a) Transformation time steps against HMM states.

(b) Transformation time steps against monitor states.

Fig. 11: Time in the transformation step compared to the size of the HMM $\mathcal{M}$ or the monitor $\mathcal{A}$ for ToVer verification.

MDP scales with the number of states in each. Table 2 contains the full results of the monitor verification experiments. The columns give the family name, an ID, learning threshold, horizon, and whether we check for missed alarms or false alarms. We give the size of the HMM (states,transitions), the number of observations, and the size of the DFA (states,transtitions). Furthermore, we list the run time for the complete verification procedure as well as the time spent on transforming the problem into a policy synthesis problem and the policy sythesis in PAYNT. Dashes in λ_l indicate that, instead of verifying a bound, we are verifying an extremum. Dashes in λ^{found} indicate no trace was found with a risk above (or below, for FA) the indicated threshold. If all values in the ToVer section of a row contain dashes, the memory limit was reached during verification.

Luko: Update with added collumns

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
AirportA-3	A-0	3/10	10	MA	45	113	18	23	414	10^{12}	1.82	$\leq 1s$	1.64	328	0.30
AirportA-3	A-1	-	10	MA	45	113	18	23	414	10^{12}	4.51	$\leq 1s$	4.32	328	1.00
AirportA-3	A-2	3/10	10	FA	45	113	18	23	414	10^{13}	6.00	$\leq 1s$	5.82	328	0.19
AirportA-3	A-3	-	10	FA	45	113	18	23	414	10^{13}	1816.83	$\leq 1s$	1816.63	328	0.01
AirportA-3	A-4	3/10	10	MA	45	113	18	51	918	10^{13}	39.29	$\leq 1s$	39.00	419	0.73
AirportA-3	A-5	-	10	MA	45	113	18	51	918	10^{13}	42.30	$\leq 1s$	42.01	419	1.00
AirportA-3	A-6	3/10	10	FA	45	113	18	51	918	10^{13}	1.23	$\leq 1s$	$\leq 1s$	419	0.26
AirportA-3	A-7	-	10	FA	45	113	18	51	918	10^{13}	6588.71	$\leq 1s$	6588.41	419	0.10
AirportA-3	A-8	3/10	10	MA	45	113	18	81	1458	10^{12}	900.48	$\leq 1s$	900.08	-	-
AirportA-3	A-9	-	10	MA	45	113	18	81	1458	10^{12}	851.17	$\leq 1s$	850.76	537	0.30
AirportA-3	A-10	3/10	10	FA	45	113	18	81	1458	10^{13}	1070.69	$\leq 1s$	1070.27	-	-
AirportA-3	A-11	-	10	FA	45	113	18	81	1458	10^{13}	1003.98	$\leq 1s$	1003.57	537	0.30
AirportA-3	A-12	3/10	10	MA	145	423	50	25	1250	10^{14}	2.05	$\leq 1s$	1.27	615	1.00
AirportA-3	A-13	-	10	MA	145	423	50	25	1250	10^{14}	2.34	$\leq 1s$	1.57	615	1.00
AirportA-3	A-14	3/10	10	FA	145	423	50	25	1250	10^{13}	9.30	$\leq 1s$	8.51	615	0.26
AirportA-3	A-15	-	10	FA	145	423	50	25	1250	10^{13}	372.71	$\leq 1s$	371.95	615	0.01
AirportA-3	A-16	3/10	10	MA	145	423	50	111	5550	10^{14}	5.90	1.97	3.89	1061	1.00
AirportA-3	A-17	-	10	MA	145	423	50	111	5550	10^{14}	5.93	1.96	3.93	1061	1.00
AirportA-3	A-18	3/10	10	FA	145	423	50	111	5550	10^{13}	4.14	1.94	2.16	1061	0.29
AirportA-3	A-19	-	10	FA	145	423	50	111	5550	10^{13}	370.22	1.96	368.23	1061	0.08
AirportA-3	A-20	3/10	10	MA	145	423	50	180	9000	10^{14}	239.02	1.94	237.05	-	-
AirportA-3	A-21	-	10	MA	145	423	50	180	9000	10^{14}	229.11	1.97	227.08	1388	0.30

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
AirportA-3	A-22	3/10	10	FA	145	423	50	180	9000	10^{13}	400.93	1.97	398.96	-	-
AirportA-3	A-23	-	10	FA	145	423	50	180	9000	10^{13}	385.53	1.99	383.51	1388	0.30
AirportA-7	A-24	3/10	10	MA	54	150	18	45	810	10^{11}	3.44	$\leq 1s$	2.99	510	0.31
AirportA-7	A-25	-	10	MA	54	150	18	45	810	10^{11}	131.15	$\leq 1s$	130.71	510	0.56
AirportA-7	A-26	3/10	10	FA	54	150	18	45	810	10^{11}	2.34	$\leq 1s$	1.91	510	0.17
AirportA-7	A-27	-	10	FA	54	150	18	45	810	10^{11}	1569.73	$\leq 1s$	1569.30	510	0.03
AirportA-7	A-28	3/10	10	MA	54	150	18	83	1494	10^{11}	1.97	$\leq 1s$	1.21	843	0.30
AirportA-7	A-29	-	10	MA	54	150	18	83	1494	10^{11}	172.46	$\leq 1s$	171.69	843	0.44
AirportA-7	A-30	3/10	10	FA	54	150	18	83	1494	10^{11}	5.53	$\leq 1s$	4.79	843	0.17
AirportA-7	A-31	-	10	FA	54	150	18	83	1494	10^{11}	960.95	$\leq 1s$	960.19	843	0.10
AirportA-7	A-32	3/10	10	MA	54	150	18	159	2862	10^{11}	330.61	1.31	329.28	-	-
AirportA-7	A-33	-	10	MA	54	150	18	159	2862	10^{11}	313.06	1.30	311.72	1107	0.30
AirportA-7	A-34	3/10	10	FA	54	150	18	159	2862	10^{11}	842.90	1.29	841.61	-	-
AirportA-7	A-35	-	10	FA	54	150	18	159	2862	10^{11}	788.76	1.31	787.43	1107	0.30
AirportA-7	A-36	3/10	10	MA	128	440	32	23	736	10^{13}	1.66	$\leq 1s$	1.19	749	0.32
AirportA-7	A-37	-	10	MA	128	440	32	23	736	10^{13}	358.82	$\leq 1s$	358.33	749	1.00
AirportA-7	A-38	3/10	10	FA	128	440	32	23	736	10^{13}	2.02	$\leq 1s$	1.58	749	0.07
AirportA-7	A-39	-	10	FA	128	440	32	23	736	10^{13}	6.37	$\leq 1s$	5.95	749	0.00
AirportA-7	A-40	3/10	10	MA	128	440	32	203	6496	10^{13}	57.63	1.24	56.36	2019	0.33
AirportA-7	A-41	-	10	MA	128	440	32	203	6496	10^{13}	541.87	1.25	540.59	2019	0.55
AirportA-7	A-42	3/10	10	FA	128	440	32	203	6496	10^{13}	4.98	1.25	3.71	2019	0.08
AirportA-7	A-43	-	10	FA	128	440	32	203	6496	10^{13}	369.47	1.17	368.28	2019	0.04
AirportA-7	A-44	3/10	10	MA	128	440	32	394	12608	10^{13}	1272.16	3.04	1269.04	-	-
AirportA-7	A-45	-	10	MA	128	440	32	394	12608	10^{13}	1242.66	3.00	1239.56	3054	0.30
AirportA-7	A-46	3/10	10	FA	128	440	32	394	12608	10^{13}	471.66	3.03	468.63	-	-
AirportA-7	A-47	-	10	FA	128	440	32	394	12608	10^{13}	449.98	3.02	446.94	3054	0.30
AirportB-3	A-48	3/10	10	MA	90	334	18	19	342	10^{13}	2.21	$\leq 1s$	1.86	679	0.97
AirportB-3	A-49	-	10	MA	90	334	18	19	342	10^{13}	10.37	$\leq 1s$	10.02	679	1.00
AirportB-3	A-50	3/10	10	FA	90	334	18	19	342	10^{13}	1.63	$\leq 1s$	1.26	679	0.11
AirportB-3	A-51	-	10	FA	90	334	18	19	342	10^{13}	12819.43	$\leq 1s$	12819.09	679	0.09
AirportB-3	A-52	3/10	10	MA	90	334	18	79	1422	10^{12}	906.65	$\leq 1s$	906.13	1071	0.45
AirportB-3	A-53	-	10	MA	90	334	18	79	1422	10^{12}	1224.41	$\leq 1s$	1223.85	1071	1.00
AirportB-3	A-54	3/10	10	FA	90	334	18	79	1422	10^{13}	2.85	$\leq 1s$	2.36	1071	0.27
AirportB-3	A-55	-	10	FA	90	334	18	79	1422	10^{13}	2004.45	$\leq 1s$	2003.96	1071	0.27
AirportB-3	A-56	3/10	10	MA	90	334	18	121	2178	10^{12}	1301.46	$\leq 1s$	1300.62	-	-
AirportB-3	A-57	-	10	MA	90	334	18	121	2178	10^{12}	1284.18	$\leq 1s$	1283.32	1535	0.30
AirportB-3	A-58	3/10	10	FA	90	334	18	121	2178	10^{13}	5921.84	$\leq 1s$	5921.01	-	-
AirportB-3	A-59	-	10	FA	90	334	18	121	2178	10^{13}	5635.54	$\leq 1s$	5634.71	1535	0.30
AirportB-3	A-60	3/10	10	MA	290	1258	50	140	7000	10^{14}	5.51	2.01	3.47	2395	0.50
AirportB-3	A-61	-	10	MA	290	1258	50	140	7000	10^{14}	5.95	1.93	3.98	2395	1.00
AirportB-3	A-62	3/10	10	FA	290	1258	50	140	7000	10^{13}	27.74	1.95	25.75	2395	0.29
AirportB-3	A-63	-	10	FA	290	1258	50	140	7000	10^{13}	1253.56	1.96	1251.56	2395	0.01
AirportB-3	A-64	3/10	10	MA	290	1258	50	147	7350	10^{14}	5.97	2.03	3.89	2433	0.40
AirportB-3	A-65	-	10	MA	290	1258	50	147	7350	10^{14}	6.48	2.08	4.36	2433	1.00
AirportB-3	A-66	3/10	10	FA	290	1258	50	147	7350	10^{13}	7.55	1.99	5.52	2433	0.29
AirportB-3	A-67	-	10	FA	290	1258	50	147	7350	10^{13}	430.78	2.06	428.67	2433	0.01
AirportB-3	A-68	3/10	10	MA	290	1258	50	244	12200	10^{14}	555.60	3.19	552.37	-	-
AirportB-3	A-69	-	10	MA	290	1258	50	244	12200	10^{14}	533.03	3.14	529.81	2933	0.30
AirportB-3	A-70	3/10	10	FA	290	1258	50	244	12200	10^{13}	847.31	3.26	844.05	-	-
AirportB-3	A-71	-	10	FA	290	1258	50	244	12200	10^{13}	820.48	3.15	817.28	2933	0.30
AirportB-7	A-72	3/10	10	MA	108	432	18	87	1566	10^{11}	108.67	1.02	107.63	1691	0.30
AirportB-7	A-73	-	10	MA	108	432	18	87	1566	10^{11}	232.95	$\leq 1s$	231.96	1691	0.84
AirportB-7	A-74	3/10	10	FA	108	432	18	87	1566	10^{11}	4.42	$\leq 1s$	3.43	1691	0.21
AirportB-7	A-75	-	10	FA	108	432	18	87	1566	10^{11}	5242.63	$\leq 1s$	5241.67	1691	0.16
AirportB-7	A-76	3/10	10	MA	108	432	18	124	2232	10^{11}	2.68	$\leq 1s$	1.83	1963	0.36

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
AirportB-7	A-77	-	10	MA	108	432	18	124	2232	10^{11}	379.94	$\leq 1s$	379.06	1963	0.53
AirportB-7	A-78	3/10	10	FA	108	432	18	124	2232	10^{11}	33.90	$\leq 1s$	33.08	1963	0.30
AirportB-7	A-79	-	10	FA	108	432	18	124	2232	10^{11}	2456.65	$\leq 1s$	2455.78	1963	0.16
AirportB-7	A-80	3/10	10	MA	108	432	18	166	2988	10^{11}	543.64	1.36	542.26	-	-
AirportB-7	A-81	-	10	MA	108	432	18	166	2988	10^{11}	514.19	1.33	512.83	2425	0.30
AirportB-7	A-82	3/10	10	FA	108	432	18	166	2988	10^{11}	2246.52	1.28	2245.23	-	-
AirportB-7	A-83	-	10	FA	108	432	18	166	2988	10^{11}	2162.99	1.29	2161.69	2425	0.30
AirportB-7	A-84	3/10	10	MA	256	1240	32	74	2368	10^{13}	5.47	1.28	4.17	2219	0.30
AirportB-7	A-85	-	10	MA	256	1240	32	74	2368	10^{13}	605.58	1.35	604.19	2219	0.96
AirportB-7	A-86	3/10	10	FA	256	1240	32	74	2368	10^{13}	4.71	1.30	3.39	2219	0.02
AirportB-7	A-87	-	10	FA	256	1240	32	74	2368	10^{13}	325.71	1.26	324.42	2219	0.00
AirportB-7	A-88	3/10	10	MA	256	1240	32	208	6656	10^{13}	431.45	2.70	428.71	4095	0.31
AirportB-7	A-89	-	10	MA	256	1240	32	208	6656	10^{13}	1004.73	2.71	1001.98	4095	0.39
AirportB-7	A-90	3/10	10	FA	256	1240	32	208	6656	10^{13}	9.19	2.83	6.33	4095	0.16
AirportB-7	A-91	-	10	FA	256	1240	32	208	6656	10^{13}	568.52	2.78	565.72	4095	0.06
AirportB-7	A-92	3/10	10	MA	256	1240	32	418	13376	10^{13}	1153.30	4.63	1148.59	-	-
AirportB-7	A-93	-	10	MA	256	1240	32	418	13376	10^{13}	1130.08	4.77	1125.21	5377	0.30
AirportB-7	A-94	3/10	10	FA	256	1240	32	418	13376	10^{13}	736.69	4.62	732.07	-	-
AirportB-7	A-95	-	10	FA	256	1240	32	418	13376	10^{13}	709.01	4.77	704.21	5377	0.30
Evade	E-0	3/10	8	MA	385	1473	325	70	22750	10^{11}	25.36	20.59	3.70	370	1.00
Evade	E-1	-	8	MA	385	1473	325	70	22750	10^{11}	25.61	20.84	3.72	370	1.00
Evade	E-2	3/10	8	FA	385	1473	325	70	22750	10^{10}	24.40	20.66	3.74	-	-
Evade	E-3	-	8	FA	385	1473	325	70	22750	10^{10}	25.49	20.82	3.77	370	0.30
Evade	E-4	3/10	8	MA	385	1473	325	132	42900	10^{11}	35.57	30.70	3.43	356	1.00
Evade	E-5	-	8	MA	385	1473	325	132	42900	10^{11}	35.25	30.54	3.38	356	1.00
Evade	E-6	3/10	8	FA	385	1473	325	132	42900	10^{10}	34.51	30.99	3.51	-	-
Evade	E-7	-	8	FA	385	1473	325	132	42900	10^{10}	35.74	30.68	3.97	356	0.30
Evade	E-8	3/10	8	MA	385	1473	325	199	64675	10^{11}	46.13	42.25	3.79	-	-
Evade	E-9	-	8	MA	385	1473	325	199	64675	10^{11}	47.59	41.89	4.21	367	0.24
Evade	E-10	3/10	8	FA	385	1473	325	199	64675	10^{10}	44.67	40.98	3.69	-	-
Evade	E-11	-	8	FA	385	1473	325	199	64675	10^{10}	46.79	41.03	4.25	367	0.30
Evade	E-12	3/10	9	MA	385	1473	325	93	30225	10^{14}	44.15	33.63	9.04	614	1.00
Evade	E-13	-	9	MA	385	1473	325	93	30225	10^{14}	44.62	33.95	9.10	614	1.00
Evade	E-14	3/10	9	FA	385	1473	325	93	30225	10^{14}	43.16	34.11	9.05	-	-
Evade	E-15	-	9	FA	385	1473	325	93	30225	10^{14}	44.81	34.29	9.04	614	0.30
Evade	E-16	3/10	9	MA	385	1473	325	167	54275	10^{14}	60.20	49.31	8.45	629	1.00
Evade	E-17	-	9	MA	385	1473	325	167	54275	10^{14}	61.74	50.75	8.53	629	1.00
Evade	E-18	3/10	9	FA	385	1473	325	167	54275	10^{14}	58.60	50.12	8.48	-	-
Evade	E-19	-	9	FA	385	1473	325	167	54275	10^{14}	60.57	49.56	8.44	629	0.30
Evade	E-20	3/10	9	MA	385	1473	325	288	93600	10^{14}	79.85	70.47	9.24	-	-
Evade	E-21	-	9	MA	385	1473	325	288	93600	10^{14}	82.69	69.91	9.11	683	0.27
Evade	E-22	3/10	9	FA	385	1473	325	288	93600	10^{14}	79.30	70.28	9.01	-	-
Evade	E-23	-	9	FA	385	1473	325	288	93600	10^{14}	83.30	70.86	8.94	683	0.30
Hidden-Incen.	H-0	3/10	10	MA	397	1649	100	110	11000	10^{18}	11.70	6.17	5.39	1284	1.00
Hidden-Incen.	H-1	-	10	MA	397	1649	100	110	11000	10^{18}	12.49	6.32	6.03	1284	1.00
Hidden-Incen.	H-2	3/10	10	FA	397	1649	100	110	11000	10^{18}	14.17	6.34	7.71	1284	0.22
Hidden-Incen.	H-3	-	10	FA	397	1649	100	110	11000	10^{18}	1857.54	6.19	1851.27	1284	0.21
Hidden-Incen.	H-4	3/10	10	MA	397	1649	100	168	16800	10^{18}	10.98	7.48	3.35	1303	1.00
Hidden-Incen.	H-5	-	10	MA	397	1649	100	168	16800	10^{18}	11.46	7.89	3.42	1303	1.00
Hidden-Incen.	H-6	3/10	10	FA	397	1649	100	168	16800	10^{18}	15.20	7.79	7.28	1303	0.25
Hidden-Incen.	H-7	-	10	FA	397	1649	100	168	16800	10^{18}	3326.33	7.57	3318.68	1303	0.24
Hidden-Incen.	H-8	3/10	10	MA	397	1649	100	257	25700	10^{18}	1233.95	10.10	1223.79	-	-
Hidden-Incen.	H-9	-	10	MA	397	1649	100	257	25700	10^{18}	1310.78	9.93	1300.70	1307	0.28
Hidden-Incen.	H-10	3/10	10	FA	397	1649	100	257	25700	10^{18}	4888.19	10.27	4877.91	-	-
Hidden-Incen.	H-11	-	10	FA	397	1649	100	257	25700	10^{18}	6304.48	10.17	6294.23	1307	0.30

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
Icy-Driving	I-0	3/10	10	MA	3	6	2	2	4	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-1	-	10	MA	3	6	2	2	4	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	29	0.10
Icy-Driving	I-2	3/10	10	FA	3	6	2	2	4	10^4	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-3	-	10	FA	3	6	2	2	4	10^4	$\leq 1s$	$\leq 1s$	$\leq 1s$	29	0.33
Icy-Driving	I-4	3/10	10	MA	3	6	2	2	4	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-5	-	10	MA	3	6	2	2	4	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	29	0.10
Icy-Driving	I-6	3/10	10	FA	3	6	2	2	4	10^4	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-7	-	10	FA	3	6	2	2	4	10^4	$\leq 1s$	$\leq 1s$	$\leq 1s$	29	0.33
Icy-Driving	I-8	3/10	10	MA	3	6	2	2	4	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-9	-	10	MA	3	6	2	2	4	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	29	0.10
Icy-Driving	I-10	3/10	10	FA	3	6	2	2	4	10^4	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-11	-	10	FA	3	6	2	2	4	10^4	$\leq 1s$	$\leq 1s$	$\leq 1s$	29	0.33
Icy-Driving	I-12	3/10	25	MA	3	6	2	2	4	10^{14}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-13	-	25	MA	3	6	2	2	4	10^{14}	$\leq 1s$	$\leq 1s$	$\leq 1s$	74	0.10
Icy-Driving	I-14	3/10	25	FA	3	6	2	2	4	10^{11}	1393.58	$\leq 1s$	1393.57	-	-
Icy-Driving	I-15	-	25	FA	3	6	2	2	4	10^{11}	1186.97	$\leq 1s$	1186.95	74	0.33
Icy-Driving	I-16	3/10	25	MA	3	6	2	2	4	10^{14}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-17	-	25	MA	3	6	2	2	4	10^{14}	$\leq 1s$	$\leq 1s$	$\leq 1s$	74	0.10
Icy-Driving	I-18	3/10	25	FA	3	6	2	2	4	10^{11}	1329.78	$\leq 1s$	1329.77	-	-
Icy-Driving	I-19	-	25	FA	3	6	2	2	4	10^{11}	1211.54	$\leq 1s$	1211.52	74	0.33
Icy-Driving	I-20	3/10	25	MA	3	6	2	2	4	10^{14}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-21	-	25	MA	3	6	2	2	4	10^{14}	$\leq 1s$	$\leq 1s$	$\leq 1s$	74	0.10
Icy-Driving	I-22	3/10	25	FA	3	6	2	2	4	10^{11}	1251.53	$\leq 1s$	1251.51	-	-
Icy-Driving	I-23	-	25	FA	3	6	2	2	4	10^{11}	1286.32	$\leq 1s$	1286.31	74	0.33
Icy-Driving	I-24	3/10	3	MA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-25	-	3	MA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	8	0.10
Icy-Driving	I-26	3/10	3	FA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-27	-	3	FA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	8	0.33
Icy-Driving	I-28	3/10	3	MA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-29	-	3	MA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	8	0.10
Icy-Driving	I-30	3/10	3	FA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-31	-	3	FA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	8	0.33
Icy-Driving	I-32	3/10	3	MA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-33	-	3	MA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	8	0.10
Icy-Driving	I-34	3/10	3	FA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-35	-	3	FA	3	6	2	8	16	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	8	0.33
Icy-Driving	I-36	3/10	10	MA	27	125	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-37	-	10	MA	27	125	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	117	0.30
Icy-Driving	I-38	3/10	10	FA	27	125	2	5	10	10^5	97.96	$\leq 1s$	97.95	-	-
Icy-Driving	I-39	-	10	FA	27	125	2	5	10	10^5	92.28	$\leq 1s$	92.26	117	0.42
Icy-Driving	I-40	3/10	10	MA	27	125	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-41	-	10	MA	27	125	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	117	0.30
Icy-Driving	I-42	3/10	10	FA	27	125	2	5	10	10^5	99.31	$\leq 1s$	99.29	-	-
Icy-Driving	I-43	-	10	FA	27	125	2	5	10	10^5	92.41	$\leq 1s$	92.38	117	0.42
Icy-Driving	I-44	3/10	10	MA	27	125	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-45	-	10	MA	27	125	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	117	0.30
Icy-Driving	I-46	3/10	10	FA	27	125	2	5	10	10^5	98.84	$\leq 1s$	98.82	-	-
Icy-Driving	I-47	-	10	FA	27	125	2	5	10	10^5	92.18	$\leq 1s$	92.15	117	0.42
Icy-Driving	I-48	3/10	3	MA	27	125	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-49	-	3	MA	27	125	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	9	0.19
Icy-Driving	I-50	3/10	3	FA	27	125	2	6	12	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-51	-	3	FA	27	125	2	6	12	10^0	-	-	-	-	-
Icy-Driving	I-52	3/10	3	MA	27	125	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-53	-	3	MA	27	125	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	9	0.19
Icy-Driving	I-54	3/10	3	FA	27	125	2	6	12	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-55	-	3	FA	27	125	2	6	12	10^0	-	-	-	-	-

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
Icy-Driving	I-56	3/10	3	MA	27	125	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-57	-	3	MA	27	125	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	9	0.19
Icy-Driving	I-58	3/10	3	FA	27	125	2	6	12	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-59	-	3	FA	27	125	2	6	12	10^0	-	-	-	-	-
Icy-Driving	I-60	3/10	10	MA	52	250	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-61	-	10	MA	52	250	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	117	0.28
Icy-Driving	I-62	3/10	10	FA	52	250	2	5	10	10^5	100.12	$\leq 1s$	100.10	-	-
Icy-Driving	I-63	-	10	FA	52	250	2	5	10	10^5	93.61	$\leq 1s$	93.58	117	0.39
Icy-Driving	I-64	3/10	10	MA	52	250	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-65	-	10	MA	52	250	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	117	0.28
Icy-Driving	I-66	3/10	10	FA	52	250	2	5	10	10^5	99.98	$\leq 1s$	99.97	-	-
Icy-Driving	I-67	-	10	FA	52	250	2	5	10	10^5	92.08	$\leq 1s$	92.05	117	0.39
Icy-Driving	I-68	3/10	10	MA	52	250	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-69	-	10	MA	52	250	2	5	10	10^5	$\leq 1s$	$\leq 1s$	$\leq 1s$	117	0.28
Icy-Driving	I-70	3/10	10	FA	52	250	2	5	10	10^5	99.60	$\leq 1s$	99.58	-	-
Icy-Driving	I-71	-	10	FA	52	250	2	5	10	10^5	93.93	$\leq 1s$	93.90	117	0.39
Icy-Driving	I-72	3/10	3	MA	52	250	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-73	-	3	MA	52	250	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	9	0.19
Icy-Driving	I-74	3/10	3	FA	52	250	2	6	12	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-75	-	3	FA	52	250	2	6	12	10^0	-	-	-	-	-
Icy-Driving	I-76	3/10	3	MA	52	250	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-77	-	3	MA	52	250	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	9	0.19
Icy-Driving	I-78	3/10	3	FA	52	250	2	6	12	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-79	-	3	FA	52	250	2	6	12	10^0	-	-	-	-	-
Icy-Driving	I-80	3/10	3	MA	52	250	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-81	-	3	MA	52	250	2	6	12	10^1	$\leq 1s$	$\leq 1s$	$\leq 1s$	9	0.19
Icy-Driving	I-82	3/10	3	FA	52	250	2	6	12	10^0	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
Icy-Driving	I-83	-	3	FA	52	250	2	6	12	10^0	-	-	-	-	-
Refuel	R-0	3/10	10	MA	87	871	63	25	1575	10^{18}	2.86	1.16	1.66	517	1.00
Refuel	R-1	-	10	MA	87	871	63	25	1575	10^{18}	2.97	1.18	1.74	517	1.00
Refuel	R-2	3/10	10	FA	87	871	63	25	1575	10^{18}	2.70	1.21	1.49	-	-
Refuel	R-3	-	10	FA	87	871	63	25	1575	10^{18}	10.26	1.17	9.08	517	0.54
Refuel	R-4	3/10	10	MA	87	871	63	25	1575	10^{18}	2.88	1.17	1.66	517	1.00
Refuel	R-5	-	10	MA	87	871	63	25	1575	10^{18}	2.91	1.16	1.71	517	1.00
Refuel	R-6	3/10	10	FA	87	871	63	25	1575	10^{18}	2.66	1.17	1.48	-	-
Refuel	R-7	-	10	FA	87	871	63	25	1575	10^{18}	10.31	1.17	9.12	517	0.54
Refuel	R-8	3/10	10	MA	87	871	63	28	1764	10^{17}	2.52	1.11	1.41	-	-
Refuel	R-9	-	10	MA	87	871	63	28	1764	10^{17}	2.53	1.15	1.36	510	0.19
Refuel	R-10	3/10	10	FA	87	871	63	28	1764	10^{16}	2.71	1.18	1.53	-	-
Refuel	R-11	-	10	FA	87	871	63	28	1764	10^{16}	10.19	1.14	9.03	510	0.54
Refuel	R-12	3/10	10	MA	132	1798	72	30	2160	10^{22}	6.99	2.23	4.70	915	1.00
Refuel	R-13	-	10	MA	132	1798	72	30	2160	10^{22}	7.50	2.19	5.24	915	1.00
Refuel	R-14	3/10	10	FA	132	1798	72	30	2160	10^{21}	5.58	2.17	3.41	-	-
Refuel	R-15	-	10	FA	132	1798	72	30	2160	10^{21}	52.78	2.18	50.58	915	0.32
Refuel	R-16	3/10	10	MA	132	1798	72	30	2160	10^{22}	6.98	2.16	4.76	915	1.00
Refuel	R-17	-	10	MA	132	1798	72	30	2160	10^{22}	7.58	2.19	5.34	915	1.00
Refuel	R-18	3/10	10	FA	132	1798	72	30	2160	10^{21}	5.63	2.22	3.41	-	-
Refuel	R-19	-	10	FA	132	1798	72	30	2160	10^{21}	52.91	2.18	50.71	915	0.32
Refuel	R-20	3/10	10	MA	132	1798	72	31	2232	10^{22}	5.65	2.32	3.33	-	-
Refuel	R-21	-	10	MA	132	1798	72	31	2232	10^{22}	5.63	2.25	3.36	905	0.14
Refuel	R-22	3/10	10	FA	132	1798	72	31	2232	10^{21}	5.79	2.36	3.43	-	-
Refuel	R-23	-	10	FA	132	1798	72	31	2232	10^{21}	52.42	2.36	50.04	905	0.32
Refuel	R-24	3/10	10	MA	139	1628	107	36	3852	10^{17}	9.46	3.86	5.48	709	1.00
Refuel	R-25	-	10	MA	139	1628	107	36	3852	10^{17}	10.04	3.99	5.93	709	1.00
Refuel	R-26	3/10	10	FA	139	1628	107	36	3852	10^{16}	9.12	3.94	5.18	-	-
Refuel	R-27	-	10	FA	139	1628	107	36	3852	10^{16}	25.27	3.94	21.23	709	0.54

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
Refuel	R-28	3/10	10	MA	139	1628	107	36	3852	10^{17}	9.44	3.77	5.55	709	1.00
Refuel	R-29	-	10	MA	139	1628	107	36	3852	10^{17}	9.85	3.83	5.90	709	1.00
Refuel	R-30	3/10	10	FA	139	1628	107	36	3852	10^{16}	9.02	3.89	5.12	-	-
Refuel	R-31	-	10	FA	139	1628	107	36	3852	10^{16}	25.63	4.03	21.49	709	0.54
Refuel	R-32	3/10	10	MA	139	1628	107	39	4173	10^{17}	8.97	3.98	4.98	-	-
Refuel	R-33	-	10	MA	139	1628	107	39	4173	10^{17}	9.12	4.12	4.99	717	0.19
Refuel	R-34	3/10	10	FA	139	1628	107	39	4173	10^{16}	9.19	4.02	5.16	-	-
Refuel	R-35	-	10	FA	139	1628	107	39	4173	10^{16}	26.07	4.27	21.70	717	0.54
RefuelB	R-36	3/10	10	MA	173	3429	63	25	1575	10^{18}	5.78	1.85	3.89	1030	1.00
RefuelB	R-37	-	10	MA	173	3429	63	25	1575	10^{18}	6.26	1.89	4.32	1030	1.00
RefuelB	R-38	3/10	10	FA	173	3429	63	25	1575	10^{18}	4.91	1.90	3.01	-	-
RefuelB	R-39	-	10	FA	173	3429	63	25	1575	10^{18}	23.65	2.02	21.51	1030	0.54
RefuelB	R-40	3/10	10	MA	173	3429	63	25	1575	10^{18}	5.69	1.87	3.77	1030	1.00
RefuelB	R-41	-	10	MA	173	3429	63	25	1575	10^{18}	6.05	1.89	4.11	1030	1.00
RefuelB	R-42	3/10	10	FA	173	3429	63	25	1575	10^{18}	4.91	1.90	3.00	-	-
RefuelB	R-43	-	10	FA	173	3429	63	25	1575	10^{18}	23.80	1.99	21.69	1030	0.54
RefuelB	R-44	3/10	10	MA	173	3429	63	28	1764	10^{17}	4.60	1.82	2.77	-	-
RefuelB	R-45	-	10	MA	173	3429	63	28	1764	10^{17}	4.70	1.87	2.82	1016	0.19
RefuelB	R-46	3/10	10	FA	173	3429	63	28	1764	10^{16}	4.91	1.89	3.02	-	-
RefuelB	R-47	-	10	FA	173	3429	63	28	1764	10^{16}	23.36	1.80	21.44	1016	0.54
RefuelB	R-48	3/10	10	MA	263	7127	72	30	2160	10^{22}	14.05	4.04	9.95	1838	1.00
RefuelB	R-49	-	10	MA	263	7127	72	30	2160	10^{22}	16.02	4.08	11.88	1838	1.00
RefuelB	R-50	3/10	10	FA	263	7127	72	30	2160	10^{21}	11.06	4.14	6.93	-	-
RefuelB	R-51	-	10	FA	263	7127	72	30	2160	10^{21}	112.42	4.14	108.26	1838	0.32
RefuelB	R-52	3/10	10	MA	263	7127	72	30	2160	10^{22}	14.09	4.07	9.96	1838	1.00
RefuelB	R-53	-	10	MA	263	7127	72	30	2160	10^{22}	16.01	4.11	11.83	1838	1.00
RefuelB	R-54	3/10	10	FA	263	7127	72	30	2160	10^{21}	10.88	3.98	6.90	-	-
RefuelB	R-55	-	10	FA	263	7127	72	30	2160	10^{21}	109.90	3.98	105.90	1838	0.32
RefuelB	R-56	3/10	10	MA	263	7127	72	31	2232	10^{22}	10.78	4.00	6.77	-	-
RefuelB	R-57	-	10	MA	263	7127	72	31	2232	10^{22}	10.92	4.16	6.74	1818	0.14
RefuelB	R-58	3/10	10	FA	263	7127	72	31	2232	10^{21}	10.92	4.03	6.89	-	-
RefuelB	R-59	-	10	FA	263	7127	72	31	2232	10^{21}	110.89	4.19	106.68	1818	0.32
RefuelB	R-60	3/10	10	MA	277	6415	107	39	4173	10^{17}	17.72	5.64	11.96	1438	1.00
RefuelB	R-61	-	10	MA	277	6415	107	39	4173	10^{17}	18.71	5.54	13.05	1438	1.00
RefuelB	R-62	3/10	10	FA	277	6415	107	39	4173	10^{16}	16.09	5.73	10.36	-	-
RefuelB	R-63	-	10	FA	277	6415	107	39	4173	10^{16}	51.23	5.90	45.13	1438	0.54
RefuelB	R-64	3/10	10	MA	277	6415	107	39	4173	10^{17}	17.95	5.86	11.97	1438	1.00
RefuelB	R-65	-	10	MA	277	6415	107	39	4173	10^{17}	18.62	5.65	12.86	1438	1.00
RefuelB	R-66	3/10	10	FA	277	6415	107	39	4173	10^{16}	15.84	5.60	10.24	-	-
RefuelB	R-67	-	10	FA	277	6415	107	39	4173	10^{16}	50.82	5.60	45.03	1438	0.54
RefuelB	R-68	3/10	10	MA	277	6415	107	41	4387	10^{17}	15.80	5.83	9.95	-	-
RefuelB	R-69	-	10	MA	277	6415	107	41	4387	10^{17}	15.75	5.75	9.99	1442	0.19
RefuelB	R-70	3/10	10	FA	277	6415	107	41	4387	10^{16}	16.27	5.88	10.39	-	-
RefuelB	R-71	-	10	FA	277	6415	107	41	4387	10^{16}	50.79	5.59	45.00	1442	0.54
SnL	S-0	3/10	10	MA	101	502	4	17	68	10^{5}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
SnL	S-1	-	10	MA	101	502	4	17	68	10^{5}	2.44	$\leq 1s$	2.33	730	0.21
SnL	S-2	3/10	10	FA	101	502	4	17	68	10^{4}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
SnL	S-3	-	10	FA	101	502	4	17	68	10^{4}	$\leq 1s$	$\leq 1s$	$\leq 1s$	730	0.85
SnL	S-4	3/10	10	MA	101	502	4	17	68	10^{5}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
SnL	S-5	-	10	MA	101	502	4	17	68	10^{5}	2.45	$\leq 1s$	2.33	730	0.21
SnL	S-6	3/10	10	FA	101	502	4	17	68	10^{4}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
SnL	S-7	-	10	FA	101	502	4	17	68	10^{4}	$\leq 1s$	$\leq 1s$	$\leq 1s$	730	0.85
SnL	S-8	3/10	10	MA	101	502	4	17	68	10^{5}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
SnL	S-9	-	10	MA	101	502	4	17	68	10^{5}	2.44	$\leq 1s$	2.32	730	0.21
SnL	S-10	3/10	10	FA	101	502	4	17	68	10^{4}	$\leq 1s$	$\leq 1s$	$\leq 1s$	-	-
SnL	S-11	-	10	FA	101	502	4	17	68	10^{4}	$\leq 1s$	$\leq 1s$	$\leq 1s$	730	0.85

Table 2: Table of all verification experiments.

		Benchmark									ToVer				
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	$\lvert \mathcal{L}(\mathcal{M})^{\leq h}\rvert$	Time (s)	Trans (s)	PAYNT (s)	$\lvert \mathcal{M}_{\geqslant h}\rvert$	λ^{found}
SnL	S-12	3/10	12	MA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-13	-	12	MA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-14	3/10	12	FA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-15	-	12	FA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-16	3/10	12	MA	101	502	4	46	184	10^7	17.65	$\leq 1s$	17.39	-	-
SnL	S-17	-	12	MA	101	502	4	46	184	10^7	25.57	$\leq 1s$	25.30	1778	0.29
SnL	S-18	3/10	12	FA	101	502	4	46	184	10^6	1.64	$\leq 1s$	1.39	1778	0.00
SnL	S-19	-	12	FA	101	502	4	46	184	10^6	1.65	$\leq 1s$	1.39	1778	0.00
SnL	S-20	3/10	12	MA	101	502	4	78	312	10^7	22.01	$\leq 1s$	21.53	-	-
SnL	S-21	-	12	MA	101	502	4	78	312	10^7	30.99	$\leq 1s$	30.51	2198	0.29
SnL	S-22	3/10	12	FA	101	502	4	78	312	10^6	3.67	$\leq 1s$	3.26	-	-
SnL	S-23	-	12	FA	101	502	4	78	312	10^6	3.50	$\leq 1s$	3.06	2198	0.39
SnL	S-24	3/10	14	MA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-25	-	14	MA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-26	3/10	14	FA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-27	-	14	FA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-28	3/10	14	MA	101	502	4	139	556	10^9	11.80	1.04	10.73	6900	0.82
SnL	S-29	-	14	MA	101	502	4	139	556	10^9	159.60	1.05	158.55	6900	1.00
SnL	S-30	3/10	14	FA	101	502	4	139	556	10^8	37.77	1.04	36.71	6900	0.29
SnL	S-31	-	14	FA	101	502	4	139	556	10^8	109.64	$\leq 1s$	108.65	6900	0.14
SnL	S-32	3/10	14	MA	101	502	4	214	856	10^9	134.47	1.51	132.94	-	-
SnL	S-33	-	14	MA	101	502	4	214	856	10^9	153.42	1.77	151.60	4842	0.29
SnL	S-34	3/10	14	FA	101	502	4	214	856	10^8	233.59	1.47	232.12	-	-
SnL	S-35	-	14	FA	101	502	4	214	856	10^8	227.03	1.54	225.48	4842	0.31
SnL	S-36	3/10	16	MA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-37	-	16	MA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-38	3/10	16	FA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-39	-	16	FA	101	502	4	1	4	-	-	-	-	-	-
SnL	S-40	3/10	16	MA	101	502	4	336	1344	10^{11}	54.10	2.37	51.70	13704	0.38
SnL	S-41	-	16	MA	101	502	4	336	1344	10^{11}	319.62	2.27	317.34	13704	1.00
SnL	S-42	3/10	16	FA	101	502	4	336	1344	10^{10}	223.18	2.10	221.04	13704	0.27
SnL	S-43	-	16	FA	101	502	4	336	1344	10^{10}	672.55	2.33	670.20	13704	0.15
SnL	S-44	3/10	16	MA	101	502	4	489	1956	10^{11}	1385.82	8.08	1377.64	-	-
SnL	S-45	-	16	MA	101	502	4	489	1956	10^{11}	1317.26	8.27	1308.87	14710	0.29
SnL	S-46	3/10	16	FA	101	502	4	489	1956	10^{10}	4697.95	8.01	4689.94	-	-
SnL	S-47	-	16	FA	101	502	4	489	1956	10^{10}	4461.63	8.18	4453.43	14710	0.30

D Results from Learning Experiments

We present the complete results for the Learning experiments. Figure 12 contains the division of the runtime during ToVer learning into its component parts. We split the runtime in the following sections:

PAYNT Time spent by PAYNT verifying colored MDPS

L⋆ Time spent by the L⋆ algorithm creating a hypothesis. Does include time spent on MQs.

Transformation Time spent transforming HMMs and monitors into colored MDPs for PAYNT.

Conformance Time spent on conformance testing during the EQs.

Other Any time not spent by the above processes.

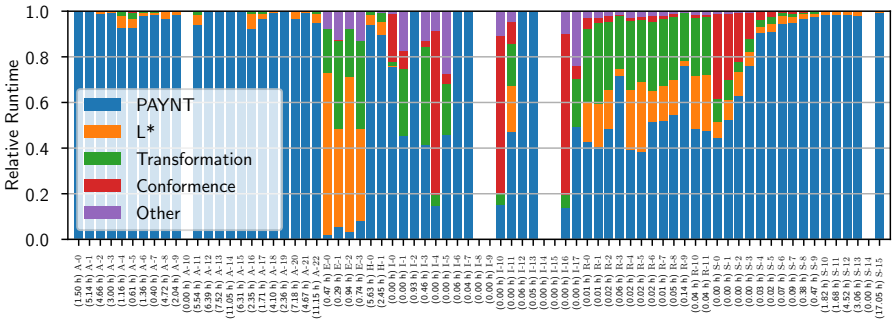


Fig. 12: Runtime division over steps in ToVer learning. Empty bars are HMMs for which learning did not finish.

Fig. 13: Legend of symbols used in plots

Figure 13 contains the symbol legend for all learning and verification experiment figures.

Table 3 contains the full results of the learning experiments. The columns give the family name and ID. They also list the threshold parameters and the horizon. We report the size of the HMM (states, transitions, number of observations). Furthermore, we detail the results of ToVer learning. We give the total runtime for the learning procedure. We also show the amount of learning rounds needed and the number of states in the learned monitor. Additionally, we present the found minimum risk of a trace accepted by the monitor and the maximum risk of a trace not accepted by the monitor. Lastly, we detail the results of the baseline learning method. We again list the amount of time spent, the number of states in the learned monitor, and the minimum and maximum threshold as before.

Whenever either the ToVer columns of a row or the baseline columns of a row only contain dashes, this method either went over the 24-hour timeout, or used more than 15 GiB of memory.

Table 3: Table of all learn experiments.

		Benchmark						ToVer					Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	Rounds	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
Airport	A-0	3/10	3/10	10	45	113	18	3540.91	26	81	0.30	0.30	230.98	165	0.30	0.30
Airport	A-1	1/10	7/20	10	45	113	18	11427.54	12	74	0.12	0.30	230.98	165	0.30	0.30
Airport	A-2	3/10	3/10	10	88	244	32	14678.85	41	209	0.30	0.30	441.39	261	0.30	1.00
Airport	A-3	1/10	7/20	10	88	244	32	7804.00	25	107	0.20	0.33	441.39	261	0.30	1.00

Table 3: Table of all learn experiments.

		Benchmark						ToVer					Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	Rounds	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
Airport	A-4	3/10	3/10	10	145	423	50	3555.83	51	180	0.30	0.30	1004.59	320	0.30	1.00
Airport	A-5	1/10	7/20	10	145	423	50	1716.58	35	120	0.10	0.34	1004.59	320	0.30	1.00
Airport	A-6	3/10	3/10	10	54	150	18	3782.46	39	159	0.30	0.30	362.28	237	0.30	0.30
Airport	A-7	1/10	7/20	10	54	150	18	941.86	12	82	0.17	0.34	362.28	237	0.30	0.30
Airport	A-8	3/10	3/10	10	128	440	32	16014.17	88	394	0.30	0.30	2613.51	655	0.30	0.30
Airport	A-9	1/10	7/20	10	128	440	32	6050.52	27	156	0.13	0.33	2613.51	655	0.30	0.30
Airport	A-10	3/10	3/10	10	235	917	50	-	-	-	-	-	-	-	-	-
Airport	A-11	1/10	7/20	10	235	917	50	18649.72	77	554	0.18	0.35	-	-	-	-
Airport	A-12	3/10	3/10	10	90	334	18	16962.93	26	121	0.30	0.30	603.35	179	0.30	0.30
Airport	A-13	1/10	7/20	10	90	334	18	17078.39	18	79	0.11	0.35	603.35	179	0.30	0.30
Airport	A-14	3/10	3/10	10	176	724	32	35225.81	43	193	0.30	0.30	681.12	263	0.30	1.00
Airport	A-15	1/10	7/20	10	176	724	32	18382.43	23	99	0.18	0.31	681.12	263	0.30	1.00
Airport	A-16	3/10	3/10	10	290	1258	50	7108.74	58	244	0.30	0.30	1638.10	313	0.30	1.00
Airport	A-17	1/10	7/20	10	290	1258	50	5025.18	31	113	0.11	0.35	1638.10	313	0.30	1.00
Airport	A-18	3/10	3/10	10	108	432	18	12450.23	35	166	0.30	0.30	616.77	237	0.30	0.30
Airport	A-19	1/10	7/20	10	108	432	18	5146.48	14	79	0.13	0.33	616.77	237	0.30	0.30
Airport	A-20	3/10	3/10	10	256	1240	32	24791.96	82	418	0.30	0.30	4103.99	673	0.30	0.30
Airport	A-21	1/10	7/20	10	256	1240	32	14360.51	26	138	0.12	0.32	4103.99	673	0.30	0.30
Airport	A-22	1/10	7/20	10	470	2550	50	37846.42	81	565	0.10	0.35	-	-	-	-
Evade	E-0	3/10	3/10	8	385	1473	325	1581.71	57	199	0.30	0.24	2405.45	251	0.30	0.24
Evade	E-1	1/10	7/20	8	385	1473	325	963.67	31	123	0.30	0.35	2405.45	251	0.30	0.24
Evade	E-2	3/10	3/10	9	385	1473	325	3209.25	79	288	0.30	0.27	4796.59	356	0.30	0.27
Evade	E-3	1/10	7/20	9	385	1473	325	2511.34	46	188	0.30	0.34	4796.59	356	0.30	0.27
Hidden-Incen.	H-0	3/10	3/10	10	397	1649	100	12711.83	62	257	0.30	0.28	1406.26	353	0.30	0.28
Hidden-Incen.	H-1	1/10	7/20	10	397	1649	100	6062.39	47	226	0.17	0.34	1406.26	353	0.30	0.28
Icy-Driving	I-0	3/10	3/10	10	3	6	2	$\leq 1s$	1	2	0.33	0.10	74.68	2	0.33	0.10
Icy-Driving	I-1	1/10	7/20	10	3	6	2	$\leq 1s$	1	2	0.33	0.10	74.68	2	0.33	0.10
Icy-Driving	I-2	3/10	3/10	25	3	6	2	1664.96	1	2	0.33	0.10	176.36	2	0.33	0.10
Icy-Driving	I-3	1/10	7/20	25	3	6	2	$\leq 1s$	1	2	0.33	0.10	176.36	2	0.33	0.10
Icy-Driving	I-4	3/10	3/10	3	3	6	2	$\leq 1s$	2	8	0.33	0.10	28.19	2	0.33	0.10
Icy-Driving	I-5	1/10	7/20	3	3	6	2	$\leq 1s$	1	2	0.33	0.10	28.19	2	0.33	0.10
Icy-Driving	I-6	3/10	3/10	10	27	125	2	105.33	2	5	0.42	0.30	93.99	5	0.42	0.30
Icy-Driving	I-7	1/10	7/20	10	27	125	2	65.12	2	5	0.42	0.30	93.99	5	0.42	0.30
Icy-Driving	I-8	3/10	3/10	25	27	125	2	-	-	-	-	-	-	-	-	-
Icy-Driving	I-9	1/10	7/20	25	27	125	2	-	-	-	-	-	-	-	-	-
Icy-Driving	I-10	3/10	3/10	3	27	125	2	$\leq 1s$	3	6	1.00	0.19	30.12	5	1.00	0.19
Icy-Driving	I-11	1/10	7/20	3	27	125	2	$\leq 1s$	2	5	1.00	0.19	30.12	5	1.00	0.19
Icy-Driving	I-12	3/10	3/10	10	52	250	2	103.20	2	5	0.39	0.28	88.25	5	0.39	0.28
Icy-Driving	I-13	1/10	7/20	10	52	250	2	92.32	2	5	0.39	0.28	88.25	5	0.39	0.28
Icy-Driving	I-14	3/10	3/10	25	52	250	2	-	-	-	-	-	-	-	-	-
Icy-Driving	I-15	1/10	7/20	25	52	250	2	-	-	-	-	-	-	-	-	-
Icy-Driving	I-16	3/10	3/10	3	52	250	2	$\leq 1s$	3	6	1.00	0.19	28.11	5	1.00	0.19
Icy-Driving	I-17	1/10	7/20	3	52	250	2	$\leq 1s$	2	5	1.00	0.19	28.11	5	1.00	0.19
Refuel	R-0	3/10	3/10	10	87	871	63	14.06	14	28	0.54	0.19	128.18	28	0.54	0.19
Refuel	R-1	1/10	7/20	10	87	871	63	15.21	17	27	0.54	0.19	128.18	28	0.54	0.19
Refuel	R-2	3/10	3/10	10	132	1798	72	23.51	16	31	0.32	0.14	119.87	31	0.32	0.14
Refuel	R-3	1/10	7/20	10	132	1798	72	154.55	18	33	0.32	0.14	119.87	31	0.32	0.14
Refuel	R-4	3/10	3/10	10	139	1628	107	39.60	20	39	0.54	0.19	141.87	42	0.54	1.00
Refuel	R-5	1/10	7/20	10	139	1628	107	43.13	26	43	0.54	0.19	141.87	42	0.54	1.00
Refuel	R-6	3/10	3/10	10	173	3429	63	26.09	15	28	0.54	0.19	122.46	28	0.54	0.19
Refuel	R-7	1/10	7/20	10	173	3429	63	25.20	16	28	0.54	0.19	122.46	28	0.54	0.19
Refuel	R-8	3/10	3/10	10	263	7127	72	42.80	18	31	0.32	0.14	139.79	33	0.32	0.14
Refuel	R-9	1/10	7/20	10	263	7127	72	387.19	18	32	0.32	0.14	139.79	33	0.32	0.14
Refuel	R-10	3/10	3/10	10	277	6415	107	65.74	21	41	0.54	0.19	154.55	40	0.54	1.00
Refuel	R-11	1/10	7/20	10	277	6415	107	69.33	23	42	0.54	0.19	154.55	40	0.54	1.00
SnL	S-0	3/10	3/10	10	101	502	4	1.98	2	17	0.85	0.21	301.11	33	0.85	0.21
SnL	S-1	1/10	7/20	10	101	502	4	3.13	2	23	0.85	0.21	301.11	33	0.85	0.21
SnL	S-2	3/10	3/10	11	101	502	4	6.06	3	42	0.40	0.21	374.93	62	0.40	0.21

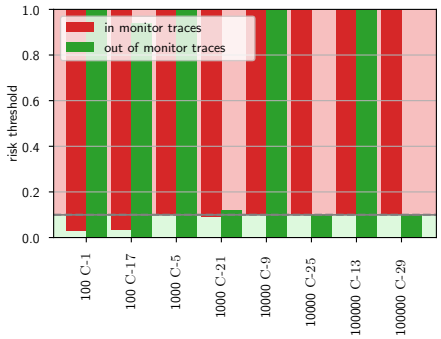
Table 3: Table of all learn experiments.

		Benchmark						ToVer					Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	Rounds	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
SnL	S-3	1/10	7/20	11	101	502	4	10.59	3	42	0.40	0.21	374.93	62	0.40	0.21
SnL	S-4	3/10	3/10	12	101	502	4	83.15	7	78	0.39	0.29	841.42	116	0.39	0.29
SnL	S-5	1/10	7/20	12	101	502	4	54.16	4	88	0.39	0.29	841.42	116	0.39	0.29
SnL	S-6	3/10	3/10	13	101	502	4	151.47	3	150	0.31	0.29	1184.96	187	0.31	0.29
SnL	S-7	1/10	7/20	13	101	502	4	203.35	6	99	0.31	0.31	1184.96	187	0.31	0.29
SnL	S-8	3/10	3/10	14	101	502	4	957.65	9	214	0.31	0.29	833.20	288	0.31	0.29
SnL	S-9	1/10	7/20	14	101	502	4	1310.91	10	215	0.31	0.32	833.20	288	0.31	0.29
SnL	S-10	3/10	3/10	15	101	502	4	4694.20	10	321	0.31	0.29	1491.02	462	0.31	0.29
SnL	S-11	1/10	7/20	15	101	502	4	4568.24	17	324	0.31	0.32	1491.02	462	0.31	0.29
SnL	S-12	3/10	3/10	16	101	502	4	10381.41	15	489	0.30	0.29	1478.43	643	0.30	0.29
SnL	S-13	1/10	7/20	16	101	502	4	7103.73	15	534	0.30	0.31	1478.43	643	0.30	0.29
SnL	S-14	3/10	3/10	17	101	502	4	-	-	-	-	-	2223.82	980	0.30	0.30
SnL	S-15	1/10	7/20	17	101	502	4	36454.28	15	735	0.18	0.34	2223.82	980	0.30	0.30

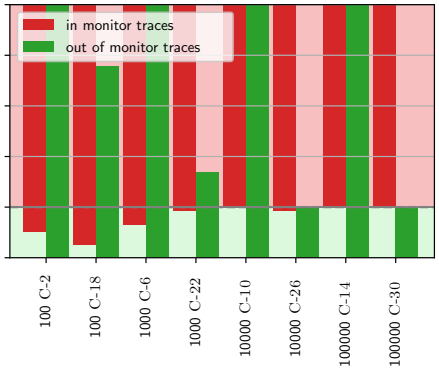
E Results from Baseline Sampling Count Experiment

We evaluate the impact of the learning threshold λ_l and the amount of samples used during conformance testing for the baseline model. Figure 14 contains the minimum risk of a trace accepted by the monitor and the maximum risk of a trace not accepted by the monitor for $\lambda_l \in \{0.05, 0.2, 0.4\}$ and sampling counts in $\{100, 1000, 10000, 100000\}$. All combinations where evaluated on AiportB-3 and AirportB-7.

Using 10000 samples and 100000 samples both had the same number of missed alarms. However, 10000 Samples had more false alarms then 100000 samples. The learning threshold seemed to have no effect on the correctness of the learned monitors.



(a) Risk threshold of 0.1



(b) Risk threshold of 0.2

(c) Risk threshold of 0.4

Fig. 14: Found risk thresholds when using baseline leaning with a different amount of samples in each EQ step. This experiment is done for three different risk thresholds.